The exponential wall: a finite-size crossover criterion for shadow-based moment estimation, with a hardware case study

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Abstract

Classical shadows estimate nonlinear functions of a quantum state, such as the moments $\mathrm{Tr}(\rho^k)$, from randomized single-copy measurements. The finite-sample variance of the resulting U-statistic estimators splits into terms of different budget order — a Hoeffding decomposition [16, 17] stated for shadows in [1] — whose two decay regimes are known [18, 20]. These analyses bound the state-dependent projection variances, and separate bounds on them do not constrain their ratio; we evaluate them in closed form for depolarized Haar-random pure states, locating the budget $M^* = \zeta_2/(2\zeta_1)$ at which the estimator's effective error exponent crosses over. Over the sizes we reach ($n = 2$ to 9) this budget grows approximately exponentially, with a fitted base near 5.3 for this ensemble — an empirical finite-size fit over that range, not a proven asymptotic rate.

Evaluating the coefficients turns the regime structure into a plug-in criterion, with no parameters fitted to the crossover data, for when a collective two-copy measurement attains a lower RMSE at equal copy budget than single-copy shadows. Across 83 simulated cells spanning four state ensembles, three of them held out from the derivation, the criterion predicts the crossover size to within one qubit in 82. These cells are simulation throughout, and the collective side of the $k > 2$ cells is modelled as a bounded ± 1 outcome that does not carry the gates, depth, or routing its circuit would require.

We then confront the prediction with what the platform reports as a 108-qubit superconducting processor, with predictions committed before submission. Both failed: a Bell-state purity locked at 0.89 measured 0.72, and a single-copy anchor locked at 1.50 measured 1.34 with disjoint intervals; byte-identical circuits also varied substantially between sessions. We do not demonstrate the crossover on hardware — at $n \geq 3$ the single-copy baseline is model output, and the one point we could measure failed — but the failure is informative: static, single-session characterization did not predict collective-measurement performance on the platform and execution path we tested.

1. INTRODUCTION

At ten qubits, on the noisy-pure states we benchmark and at a fixed measurement budget, the standard single-copy method for estimating the purity of a quantum state returns an error fifteen times larger than the quantity it is estimating. The estimator is still unbiased; it is the spread that has become useless.

That failure matters because the quantity is not exotic. The moments $\mathrm{Tr}(\rho^k)$ of an unknown state are the entry point to a large fraction of what one wants to know about it. The second moment is the purity. Logarithms of the moments give the Rényi entropies. Moments of the partial transpose furnish PPT entanglement criteria and bounds on negativity. Purity appears directly in error-mitigation protocols such as virtual distillation. Any experiment that wants to characterize the state its device actually prepared, rather than the state it intended to prepare, needs these numbers.

Two routes exist to obtain them. The single-copy route measures one copy of the state at a time in randomized local bases, which builds classical shadows [1], and forms an unbiased U-statistic from the snapshots. It uses no entangling gates and is therefore fairly hardware-friendly. The collective route holds k copies simultaneously and measures a joint observable, the cyclic-permutation operator, whose expectation is exactly $\mathrm{Tr}(\rho^k)$. It has $O(1)$ variance but requires entangling operations across copies.

The asymptotic separation between the two is settled. Single-copy strategies require exponentially more samples than collective ones for a family of learning tasks [2, 3], and, for protocols restricted to an identical single-copy projection-valued measurement, the lower bound for purity estimation is $\Omega(\max\{1/\varepsilon^2, 2^{n/2}/\varepsilon\})$ [4]. What is not settled is the practical question that an experimentalist actually faces: on this device, at this noise level, with this system size, which route costs less? The collective route buys variance reduction and pays in noise-induced bias. Whether that trade is worth making depends on the numbers, and the numbers have not been computed.

The gap is visible in recent work. A May 2026 study of quantum machine learning advantage on tens of noisy qubits [5] declined to give its measure-first protocol multi-copy access, citing both a theoretical result specific to their learning task (even multiple noiseless copies do not let a measure-first protocol solve it efficiently) and the resource demands of multi-copy entangled measurements, which increase qubit count, two-qubit depth, and routing overhead. Their learning task is not moment estimation, so their open questions are not ours; but their caution about the resource cost of multi-copy measurement is a premise we can now test quantitatively. On the theory side, a recent result [6] shows that the exponential two-copy advantage for purity *testing* collapses under local depolarizing noise, and that SWAP-test and Bell-basis primitives are fragile to noise absent latent error-correcting structure; their task is testing and ours is estimation, a distinction we take up in Section 7. The caution is reasonable. What has not been available is a prediction of the system size at which it stops applying.

The finite-sample variance of a shadow-based moment estimator splits into projection terms carrying different powers of the budget: the standard Hoeffding decomposition for U-statistics [16, 17], stated for classical shadows in [1]. That the resulting error decay rate changes with the budget is established as well. Elben et al. [18] state both regimes in their main text and plot them, and Aasen and Gärttner [20] track the scaling exponent migrating between the same two rates under detector noise. What those analyses do not do is locate where the regimes meet. Bounding ζ_1 and ζ_2 separately constrains their ratio not at all, and the exact second moment that would fix it is applied in the literature to worst-case sample complexity rather than to a state.

Thesis. We evaluate the state-dependent projection variances for depolarized Haar-random pure states at a single noise rate, numerically, and feed them into the exact variance formula. That fixes the crossover budget $M^* = \zeta_2/(2\zeta_1)$, which grows exponentially in n with a base near five over the sizes we reach, and, set against two exact bias laws for the collective route, yields a prediction — with nothing fitted to the crossover data — of the system size at which a collective two-copy measurement attains a lower RMSE at equal copy budget than single-copy shadows. The same evaluation runs at $k = 3$ and 4, where prior treatments bound the higher-moment variance, Elben et al.'s Appendix D.2 among them, and where the two-term truncation of the exact formula no longer suffices even when its coefficients are correct. A hardware comparison of the two routes has been reported [19]; what we add is the predictive law and a test of it locked in advance. We confront that prediction with a superconducting device and report the outcome as measured, including a committed-in-advance prediction the hardware did not bear out. What the confrontation established is the shape of the obstacle: the dominant error was readout rather than the entangling overhead the crossover argument prices, and device parameters characterized in earlier sessions did not describe the device we ran on.

Roadmap. Section 2 fixes the estimators and shows that two seemingly innocuous choices, the tuple-sampling scheme and the state ensemble, each invert the conclusion of the benchmark if made carelessly. Section 3 derives the exact variance law from the Hoeffding decomposition, gives the exact single-qubit projection variance, and establishes the α transition together with its out-of-sample validation. Section 4 derives the two exact collective bias laws and shows that the relevant distinction is the geometry of the noise channel, not its rate. Section 5 combines them into a crossover criterion with nothing fitted to the crossover data it predicts, and validates it against 83 simulated crossover cells across four state ensembles. Section 6 reports a test with predictions committed before execution on a 108-qubit superconducting processor: the measured value falls below the prediction, and the diagnosis, which bounds or excludes seven candidate mechanisms with quantitative arguments, is the section's content. Section 7 positions the work against the recent literature, Section 8 states what the work does not establish, and Section 9 concludes.

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2. SETTING AND ESTIMATORS

2.1. Local random-unitary classical shadows

Let ρ be an unknown n-qubit state. A single snapshot is produced by drawing an independent Haar-random single-qubit unitary U_q for each qubit, applying $\bigotimes_q U_q$, measuring in the computational basis to obtain a bitstring b, and forming

$$G = \bigotimes_{q=1}^{n} \left(3\, U_q^\dagger |b_q\rangle\langle b_q| U_q - \mathbb{I}\right) .$$

The snapshot is an unbiased estimator of the state, $\mathbb{E}[G] = \rho$. A budget of M snapshots consumes M copies of ρ.

2.2. The estimator must be the exact U-statistic

The k-th moment is estimated by the U-statistic over distinct k-tuples of snapshots,

$$U_M = \binom{M}{k}^{-1} \sum_{i_1<\cdots<i_k} h(G_{i_1},\ldots,G_{i_k}), \qquad h(g_1,\ldots,g_k) = \frac{1}{k!}\sum_{\pi\in S_k} \operatorname{Re}\operatorname{Tr}\big(g_{\pi(1)}\cdots g_{\pi(k)}\big)\,,$$

which is exactly unbiased: $\mathbb{E}[U_M] = \operatorname{Tr}(\rho^k)$.

The tuples are formed exhaustively rather than sampled, which is optimal at a fixed copy budget though not mandatory in total computational cost: an incomplete U-statistic trades statistical variance for lower classical compute and memory. Forming tuples from already-collected snapshots is classical post-processing and consumes no additional copies of the state, so subsampling them saves nothing physical while inflating the variance. In our sweeps a subsampled estimator inflated the single-copy RMSE by a factor of about five at $n = 2$, growing to more than fiftyfold at $n = 4$, relative to the exact one. That factor is larger than the effect the benchmark is even trying to measure, and it is also enough to reverse the benchmark's conclusion. Any comparison between single-copy and collective measurement must use the exact estimator on the single-copy side, or it is measuring an implementation choice rather than physics.

For $k = 2$ the full U-statistic has an exact closed form, and the two ways of evaluating it trade M against n. The power-sum reduction $\operatorname{Tr}(S^2) - \sum_i \operatorname{Tr}(G_i^2)$, with $S = \sum_i G_i$, is linear in M, but it builds dense $2^n \times 2^n$ matrices and so costs $O(M4^n)$. We instead use the per-qubit factorization $\operatorname{Tr}(G_iG_j) = \prod_q \operatorname{Tr}(G_i^q G_j^q)$, which forms an $M \times M$ pairwise-trace matrix in $O(M^2 n)$ time and $O(M^2)$ memory and stays computable at the sizes we reach. For $k = 4$ no complete power-sum reduction exists, because the snapshots do not commute. We give an exact construction in Appendix A by Möbius inversion over the 15 set partitions of the four cyclic slots, with the alternating (ABAB) partition, which has no power-sum representation, evaluated by a tensor contraction. The construction matches brute-force enumeration to $\lesssim 10^{-13}$.

2.3. The collective route

Let C_k be the cyclic permutation operator on k registers of dimension $d = 2^n$. It satisfies

$$\operatorname{Tr}\big(C_k\,\rho^{\otimes k}\big) = \operatorname{Tr}(\rho^k), \qquad \operatorname{Tr}(C_k) = d,$$

the second identity because the cyclic permutation on k registers has a single cycle. The second identity is what makes the depolarizing bias law of Section 4.1 come out linear in the noise rate.

We use the destructive SWAP test rather than the ancilla-based controlled-SWAP construction, because the latter decomposes into many two-qubit gates and would otherwise be dominated by gate noise on the current hardware. Two copies of the state are prepared on $2n$ qubits. For each pair $(i, i+n)$ a CNOT is applied with control i and target $i+n$, followed by a Hadamard on qubit i, and all $2n$ qubits are measured. The purity is recovered from the bitstrings by the parity rule

$$\mathrm{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{\, i \,:\, a_i = b_i = 1\,\}}\, P(\text{bitstring}),$$

where a_i, b_i are the outcomes on the two copies of qubit i. The circuit uses exactly n two-qubit gates for an n-qubit state, requires no ancilla, and has depth two after state preparation. We verified the sign rule against exact simulation to 10^{-16} (Appendix B). The gate count is the reason the entangling overhead is affordable on real hardware (Section 6).

This construction is specific to $k = 2$. The swap operator C_2 is Hermitian with eigenvalues ± 1, so its expectation is directly a two-valued measurement and no ancilla is needed. For $k \geq 3$ the cyclic operator is not Hermitian — C_3 has the cube roots of unity as eigenvalues and C_4 has $\{1, -1, \pm i\}$ — and no analogous pairwise destructive parity test exists: its Hermitian part could in principle be measured after a basis transformation, but there is no direct ± 1 analogue of the $k = 2$ circuit. For $k \geq 3$ we therefore take the standard ancilla-based Hadamard test: an ancilla is prepared in $|+\rangle$, a controlled-C_k is applied to the k copies, and the ancilla is measured in the X basis. Its outcome is ± 1 with expectation $\mathrm{Re}\,\mathrm{Tr}(C_k \rho^{\otimes k}) = \mathrm{Tr}(\rho^k)$, so the shot variance is $1 - \mu^2$ exactly as at $k = 2$, which is the model used for all k in Section 5. The cost is not comparable: the Hadamard test requires an ancilla and a controlled cyclic shift on kn qubits, against the n two-qubit gates at depth two of the $k = 2$ destructive test. Our $k \geq 3$ results are simulated, and the gate counts reported in Section 6.2 are for $k = 2$ only.

2.4. Copy-fair accounting

A single-copy protocol at budget M consumes M copies of ρ. A k-copy collective test at budget M consumes M/k measurements, each using k copies. All comparisons in this paper hold the total number of state copies fixed.

2.5. The state ensemble is not a free choice

The variance of shadow-based purity estimation scales with the purity of the state being measured, a fact we derive exactly in Section 3.3. The consequence is a trap, though not the one it first appears to be. A random Ginibre state under heavy depolarizing noise has purity approaching 2^{-n}, which tends to zero with the system's size, and a quantity that is approximately zero is easy to estimate at a fixed absolute error — the same is not true of relative precision, nor of testing, where the gap to be resolved shrinks too (Section 7). What does not follow is that the estimator is well behaved there. Those are different claims, and only the first is true.

Low purity does not remove the exponential cost. It concentrates it. At $\rho = \mathbb{I}/2^n$ every snapshot gives $\mathrm{Tr}(G\rho) = 2^{-n}$ deterministically, so $\zeta_1 = 0$ exactly, as the single-qubit identity of Section 3.3 already records at zero Bloch length. The threshold $M^* = \zeta_2/(2\zeta_1)$ therefore diverges and the estimator sits in the higher-order regime at every budget. The kernel term does not vanish with ζ_1: the single-qubit second moment $\mathbb{E}[\mathrm{Tr}(G_1G_2)^2]$ is 7 for a maximally mixed qubit, so $\zeta_2 = 7^n - 4^{-n}$, and the variance $2\zeta_2/M(M-1)$ gives an error of order $7^{n/2}/M$ — exponential in n, against a target that is itself shrinking as 2^{-n}.

The exponential is therefore a property of the estimator rather than of high-purity states. Its base is set by ζ_2, which barely moves with purity: exactly 7 per qubit at the maximally mixed state, and 6.93 fitted on the noisy-pure ensemble below. Purity governs ζ_1, and so M^* and the exponent, not the exponential itself. The ensemble still decides what the benchmark measures, but for a different reason than an easy estimator: at $\rho = \mathbb{I}/2^n$ the collective bias vanishes identically too for every *unital* channel, because $\mathrm{Tr}(\rho^k)$ then sits exactly on the depolarizing floor $2^{n(1-k)}$ and a unital channel leaves $\mathbb{I}/2^n$ fixed, so both routes degenerate at once and the comparison stops being informative. The restriction is not decorative: amplitude damping is not unital, and it carries $\mathbb{I}/2$ to a state of purity $(1+\gamma^2)/2$.

One relevant NISQ regime, and the one we benchmark on, is noisy-pure: a prepared pure state subject to modest depolarizing noise, with purity of order one. We therefore use

$$\rho = (1-q)|\psi\rangle\langle\psi| + q\,\mathbb{I}/2^n, \qquad |\psi\rangle \sim \text{Haar}, \qquad q \in [0.05,\, 0.3],$$

for which the purity is stable across the sizes tested but is set by q, approaching $(1-q)^2$ at large n: approximately 0.90 at $q = 0.05$, 0.81 at the representative $q = 0.1$ used for the headline states, and 0.49 at $q = 0.3$. The reason to benchmark here is not that the alternative is easy. It is that these states keep the problem both hard and meaningful at once: the moment being estimated is of order one, so an absolute error is a statement about something, and the collective route has a bias to pay, so the two routes are actually in competition. Choosing the wrong ensemble does not make the estimator cheap; it makes the comparison vacuous.

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3. THE SINGLE-COPY VARIANCE LAW

3.1. The Hoeffding decomposition gives the variance exactly

For a symmetric kernel h of order k, define the c-th order projection and its variance,

$$h_c(g_1,\dots,g_c) = \mathbb{E}_{G_{c+1},\dots,G_k}[h(g_1,\dots,g_c,G_{c+1},\dots,G_k)]\,, \qquad \zeta_c = \mathrm{Var}[h_c(G_1,\dots,G_c)]\,.$$

The variance of the U-statistic is then exactly

$$\boxed{\mathrm{Var}(U_M) = \binom{M}{k}^{-1}\sum_{c=1}^{k}\binom{k}{c}\binom{M-k}{k-c}\zeta_c}$$

This is the standard Hoeffding decomposition for U-statistics [16, 17], and we claim no novelty for it; what is new here is the closed-form evaluation of the second-order ζ_c and the threshold their ratio fixes (Section 3 3.5).

For $k = 2$ this reduces to

$$\mathrm{Var}(U_M) = \frac{4(M-2)\,\zeta_1 + 2\,\zeta_2}{M(M-1)}, \qquad \zeta_1 = \mathrm{Var}[\mathrm{Tr}(G\rho)]\,, \quad \zeta_2 = \mathrm{Var}[\mathrm{Tr}(G_iG_j)]\,.$$

We verified the general formula against brute-force Monte Carlo of the exact estimator at $k = 3$ and $k = 4$, on noisy-pure, GHZ, and low-rank states, with all ratios within 3%, and confirmed that it reduces to the $k = 2$ form at a ratio of 1.000000. The law is exact, not asymptotic, and it holds at every budget.

Because the k-argument kernel is multilinear in the snapshots, every projection variance depends on the shadow ensemble only through its second-moment matrix; local Haar single-qubit shadows and local Pauli-basis shadows have identical second-moment matrices, so all results of this section hold verbatim for either.

One subtlety is easy to get wrong and can pass unnoticed. For $k = 2$ the second-order projection *is* the kernel, so ζ_2 is the kernel variance. For $k \geq 3$ this is false: ζ_2 is the two-argument projection, and the kernel variance is ζ_k. Substituting one for the other is wrong by roughly a factor of seven and corrupts any variance estimate at higher moments while appearing to work at $k = 2$. The resulting approximation breaks down at $k \geq 3$, for reasons that can look like physics but are not.

3.2. Two terms compete, and their ratio is a budget threshold

The exact variance contains two competing terms with different budget dependence. At large M the linear term dominates,

$$\mathrm{Var}(U_M) \approx \frac{4\zeta_1}{M} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1/2}, \qquad \alpha = \tfrac{1}{2},$$

which is the familiar statistical scaling that a fixed-exponent bound predicts. At small M the higher-order term dominates,

$$\mathrm{Var}(U_M) \approx \frac{2\zeta_2}{M^2} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1}, \qquad \alpha = 1.$$

The two asymptotic terms are equal at

$$\boxed{M^* = \frac{\zeta_2}{2\zeta_1}}$$

This is the balance point of the two large-M forms above; the exact numerator terms balance at a point differing from it by an additive constant, immaterial once M^* is large. Nothing that follows depends on the choice, because the predicted exponents are computed from the exact variance formula rather than from M^*. This threshold is the object the rest of the section is about. Which side of M^* a given experiment sits on determines the character of its scaling, and the projection variances that set M^* are state-dependent. To see what that dependence is, we compute them exactly for one qubit.

3.3. The single-qubit second moment, exactly

For a single qubit with density matrix r — throughout this subsection we write r for a single-qubit state, reserving ρ for the full n-qubit state, because the identity below holds only at one qubit — with Bloch length t and purity $p = (1 + t^2)/2$,

$$\boxed{\mathbb{E}\big[\mathrm{Tr}(Gr)^2\big] = \tfrac{1}{4} + \tfrac{5}{4}t^2 = \tfrac{5}{2}p - 1}$$

and the single-qubit first projection variance follows as

$$\zeta_1 = \tfrac{3}{4}t^2 - \tfrac{1}{4}t^4.$$

We verified this numerically across the full Bloch range. The identity is exact, and two consequences follow from it.

The second moment grows by a factor of six from a maximally mixed qubit, where it equals 1/4, to a pure qubit, where it equals 3/2. Compounded over n qubits, that factor is the origin of the exponential cost, and it is state-dependent rather than universal. The compounding picture is a heuristic for the factorized case: it does not carry over to arbitrary entangled or correlated states, which is what Section 3.4 shows when the weight-only ansatz fails at $n \geq 2$. This is the analytic reason why the ensemble choice of Section 2.5 is not cosmetic: a low-purity ensemble is not a harder instance of the same problem but a different one, and an easier benchmark to score well on at a fixed absolute error, because the target itself shrinks toward the depolarizing floor. The estimator's own variance stays exponential there, as Section 2.5 shows; it is the quantity being resolved that gets smaller, not the noise on it.

The identity also passes the boundary check. At $t = 0$ it gives $\zeta_1 = 0$, which is correct: for a maximally mixed qubit, $\mathrm{Tr}(Gr) = 1/2$ deterministically, with no variance to speak of.

3.4. The weight-only quadratic ansatz fails already at one qubit

The single-qubit result invites a generalization. If the second moment carries a factor of 3 per qubit in the support of a Pauli string, then ζ_1 ought to decompose over Pauli strings with a coefficient depending only on the Pauli weight,

$$\zeta_1 \stackrel{?}{=} \sum_P c_{|P|} \langle P \rangle^2.$$

This already fails at a single qubit, and provably so. With universal weight coefficients the most general such form for one qubit is $c_0\langle I\rangle^2 + c_1(\langle X\rangle^2 + \langle Y\rangle^2 + \langle Z\rangle^2) = c_0 + c_1 t^2$, an affine function of t^2, because the three squared Pauli expectations sum to the squared Bloch length t^2 and carry no further invariant. The exact single-qubit projection variance of Section 3.3 is $\zeta_1 = \frac{3}{4}t^2 - \frac{1}{4}t^4$, whose quartic term has coefficient $-\frac{1}{4}$ whatever c_0 and c_1 are chosen: no weight-only quadratic form reproduces it. The obstruction is a quartic dependence on the state that a form quadratic in the Pauli expectations cannot carry, and it is present before any question of entanglement or correlation arises.

The insufficiency persists and grows with size. Tested against a diverse family of 19 states at $n = 2$, varying depolarizing rate, rank, and structure, the best-fit weight-only ansatz leaves a residual of up to 12.3%. A weight-only quadratic form is insufficient, and Section 3 3.5 identifies exactly what it omits. The weight-only form is the diagonal of an exact cubic identity already available in [1], and the residual it leaves is that identity's off-diagonal sum. The obstruction is a quartic dependence on the state that no form quadratic in the Pauli expectations can carry, and it is present before any question of entanglement or correlation arises.

The ansatz nonetheless *appears* to hold. Tested only within a narrow single-parameter family, Haar-pure states at a fixed depolarizing rate of $q = 0.1$, it fits to 0.1%, because that family spans a low-dimensional invariant subspace. A closed form validated on one ensemble is not a closed form. At second order the projection variances follow in closed form (Section 3 3.5); at higher moment orders they are estimated numerically, and the variance law is exact in either case.

3.5. The projection variances in closed form

Section 3.4 rules out a weight-only quadratic form for ζ_1. What it does not rule out, and what holds, is a cubic one. The ingredient is already in the literature. Lemma 4 of [1] gives the exact second moment of two reconstructed Pauli coefficients under local shadows,

$$\mathbb{E}[\hat{x}_P\,\hat{x}_Q] = 3^{|\mathrm{supp}(P)\cap\mathrm{supp}(Q)|}\,\mathrm{Tr}(\rho\,P\ominus Q)\,,$$

when P and Q agree on every qubit where both are non-identity and zero otherwise, where $\hat{x}_P = \mathrm{Tr}(GP)$ and $P \ominus Q$ cancels the shared non-identity factors. We claim no novelty for this identity; what follows is its evaluation.

Since $\mathrm{Tr}(G\rho) = 2^{-n}\sum_s \langle P_s\rangle \hat{x}_s$, the first projection variance follows by contraction, and since the k-argument kernel of Section 3.1 is multilinear in the snapshots, every ζ_c is a quadratic functional of the same second-moment matrix. Two consequences precede the evaluation.

First, the diagonal $s = s'$ of that contraction is exactly the weight-only form of Section 3.4, with $c_w = 3^w/4^n$, so the residual that form leaves is exactly the off-diagonal sum. The shortfall identified there is not an open feature of the algebra.

Second, because only second moments enter, the projection variances do not distinguish local Haar single-qubit shadows from local Pauli-basis shadows: the two ensembles have entrywise identical second-moment matrices, so ζ_c agrees for every c and every k. Every statement below holds for either.

Everything in this subsection concerns $k = 2$. The higher moment orders are treated numerically as before; their projections do not reduce the same way, and we say what is known about them at the end.

a. The second projection variance is a spectral sum. The two-copy kernel is diagonal in the Pauli basis, and collecting the compatible pairs by the string they leave behind gives, for any state,

$$\zeta_2 \;=\; 7^n \sum_P \langle P\rangle^2 \left(\tfrac{1}{14}\right)^{|P|} \;-\; \mathrm{Tr}(\rho^2)^2,$$

the sum running over all 4^n Pauli strings with $|P|$ the weight. Only the identity term is state-independent, so the base of ζ_2 is never below 7, and a counting argument over compatible pairs bounds it above by 17/2. At the maximally mixed state every other term vanishes and the expression returns $7^n - 4^{-n}$, recovering Section 2.5. What sets the base is how the state's Pauli weight is distributed: a spectrum spread over all 4^n strings leaves the identity term dominant and the base at exactly 7, while a spectrum concentrated on a stabilizer group lifts it.

b. Evaluation for the noisy-pure ensemble. Averaging over the ensemble of Section 2.5, $\rho = (1-q)|\psi\rangle\langle\psi| + qI/2^n$ with $|\psi\rangle$ Haar, requires only the second and third Haar moments of Pauli expectations, $\mathbb{E}[\langle P\rangle^2] = 1/(d+1)$ and $\mathbb{E}[\langle P_u\rangle\langle P_s\rangle\langle P_{s'}\rangle] = 2/((d+1)(d+2))$ when $P_s P_{s'} = P_u$, together with four counts over compatible Pauli pairs. Writing $d = 2^n$, $u = 1-q$, and $p_2 = \mathrm{Tr}(\rho^2) = u^2 + q(2-q)/d$,

$$\zeta_1 = 4^{-n}\left[1 + \frac{u^2(10^n - 1)}{d+1} + \frac{2u^2(4^n - 1)}{d+1} + \frac{2u^3(16^n - 10^n - 2\cdot 4^n + 2)}{(d+1)(d+2)}\right] - p_2^2,$$

$$\zeta_2 = 4^{-n}\left[28^n + \frac{u^2(34^n - 28^n)}{d+1}\right] - p_2^2,$$

the second being the spectral sum above evaluated on this ensemble. Each qubit contributes one identity of weight 3^0 and three Paulis of weight 3^1, so $\sum_s 3^{|s|} = 10^n$; the compatible pairs give 16^n with that weighting and 34^n with its square, with diagonals 10^n and 28^n.

c. The threshold and what sets its base. Write β for the growth base of $\mathbb{E}[\mathrm{Tr}(G\rho)^2]$, which is also the base of ζ_1. Since $M^* = \zeta_2/(2\zeta_1)$, its base is $\mathrm{base}(\zeta_2)/\beta$ exactly, and both factors are spectral functionals of the state rather than universal constants. For the ensemble of Section 2.5 the closed forms give $\beta = 5/4$ and $\mathrm{base}(\zeta_2) = 7$, so

$$M^*(n) \;\longrightarrow\; \frac{1}{2(1-q)^2}\left(\frac{28}{5}\right)^n, \qquad \frac{28}{5} = 5.6,$$

with base and prefactor both exact.

The same two functionals can be evaluated for the other ensembles this paper uses, and they do not all agree:

state family	β	base(ζ_2)	M^* base
noisy-pure ($q = 0.1$)	5/4	7	$28/5 = 5.60$
Haar-pure	5/4	7	$28/5 = 5.60$
low-rank (rank 2)	≈ 1.21	7	≈ 5.8
product, GHZ	3/2	15/2	5.00

The last row is exact and instructive. For a pure product state the spectral sum is $\sum_P \langle P\rangle^2 14^{-|P|} = (15/14)^n$, giving $\zeta_2 = (15/2)^n - 1$, and for GHZ the stabilizer group splits into even-weight Z strings and weight-n strings, giving $\zeta_2 = \frac{1}{2}\left[(15/2)^n + (13/2)^n\right] - \frac{1}{2}$. Purity alone does not determine the base: a Haar-random pure state gives 7 and a pure product state gives 15/2. What distinguishes them is the weight profile of the Pauli spectrum, which the kernel $14^{-|P|}$ reads off. The value 28/5 is therefore the spread-spectrum case, not a universal constant, and structured stabilizer states sit on a different branch at 5.

d. Higher moment orders. The multilinearity argument above shows every ζ_c depends only on the second-moment matrix, so closed forms exist in principle at $k \geq 3$ as well. We did not obtain them: at $k = 3$ and 4 the state-dependent contributions dominate the state-independent ones, so the kernel bases are not read off the maximally mixed values, and the required counts over compatible tuples are beyond what we derived. The $k \geq 3$ coefficients used below are estimated numerically.

The fitted base reported in the next subsection drifts with the fitting window, from 5.14 over $n = 2$ to 7 to 5.60 over $n = 4$ to 9. The closed forms reproduce that drift exactly, returning 5.18, 5.36 and 5.60 over the same windows with no sampling anywhere, so the drift is a finite-size property of the fit and 28/5 is what it drifts toward.

3.6. The α transition

The projection variances for noisy-pure states, estimated over this range, give clean exponential scalings. Over $n = 2$ to 7,

$$\zeta_1 \approx 0.63 \cdot (1.35)^n, \qquad \zeta_2 \approx 1.10 \cdot (6.93)^n, \qquad M^* \approx 0.87 \cdot (5.15)^n,$$

and over the wider range $n = 2$ to 9, $M^* \approx 0.76 \cdot (5.345)^n$. The base of ζ_1 is mildly curved, reading approximately 1.35 at small n and 1.30 over the full range, which accounts for the difference between the two fits.

The kernel variance grows roughly five times faster per qubit than the linear variance. Their ratio, the threshold M^*, therefore grows approximately exponentially in n over the range we fit ($n = 2$ to 9). This is the mechanism, and within that range its consequence is direct: at a fixed budget, increasing n pushes M below M^*, the higher-order term takes over, and the effective exponent migrates from 1/2 toward 1, and past it for higher k.

Defining α as the effective exponent — the negative slope of $\log \mathrm{RMSE}$ against $\log M$ fitted by least squares over the budget range actually used, as in Appendix D, so it is a window-averaged rather than a pointwise quantity and the window itself shifts with n:

n	$M^*(n)$	predicted α	measured α
2	≈ 22	0.501	0.495 ± 0.013
4	≈ 604	0.528	0.528 ± 0.012
9	$\approx 2.3 \times 10^6$	0.998	1.006 ± 0.023

The predictions are computed from projection variances estimated from shadow snapshots, fed into the exact variance formula. Nothing is fitted to the α data (Fig. 1). Across every system size and moment order for which we have measurements, the derived law matches: 8 of 8 within two standard errors at $k = 2$, 6 of 7 at $k = 3$, where the single miss sits on the two-sigma boundary and is sensitive to the random seed, and 5 of 5 at $k = 4$.

The exact combinatorial structure is necessary, not a convenience, although $k = 2$ is the wrong place to look for the evidence. There the large-M truncation of Section 3.2, $\mathrm{Var} \approx 4\zeta_1/M + 2\zeta_2/M^2$, tracks the exact variance to within 0.05% across the budgets used here and reproduces the measured α at 8 of 8, indistinguishable from the exact law: at $k = 2$ the second projection and the full kernel variance are the same quantity, so the truncation discards nothing. They separate at higher orders. Keeping the same two terms with their correct coefficients, $k^2\zeta_1/M + \frac{1}{2}k^2(k-1)^2\zeta_2/M^2$, gives 15 of 20 across $k = 2, 3, 4$ against the exact law's 19 of 20, and misses $k = 3$, $n = 8$ by twenty-six standard errors — by then ζ_3/ζ_2 exceeds 10^4 and the discarded terms carry the variance.

The threshold base is also stable across independent Monte Carlo samplings. Derived here from $\zeta_2/(2\zeta_1)$ on a fresh estimate of the projection variances, it is 5.15 to 5.35; an earlier, independently sampled run of the same $\zeta_2/(2\zeta_1)$ construction gave 5.343. This agreement is a reproducibility check on the projection-variance estimate, not corroboration by a methodologically independent route.

A log-linear regression of $M^*(n)$ over $n = 2$ to 9 returns a base of 5.34 with a 95% confidence interval of [5.14, 5.54] and a prefactor of 0.75 ± 0.09 (regression standard errors, treating each $M^*(n)$ as fixed); ζ_1 and ζ_2 fit to bases 1.30 [1.26, 1.34] and 6.94 [6.90, 6.98]. The ζ_2 base sits just below the exact per-qubit second moment 7 at the maximally mixed state (Section 3.3) and barely moves across the fitting windows; the base drift below is carried by ζ_1. The regression residuals are not flat — positive at both ends of the range

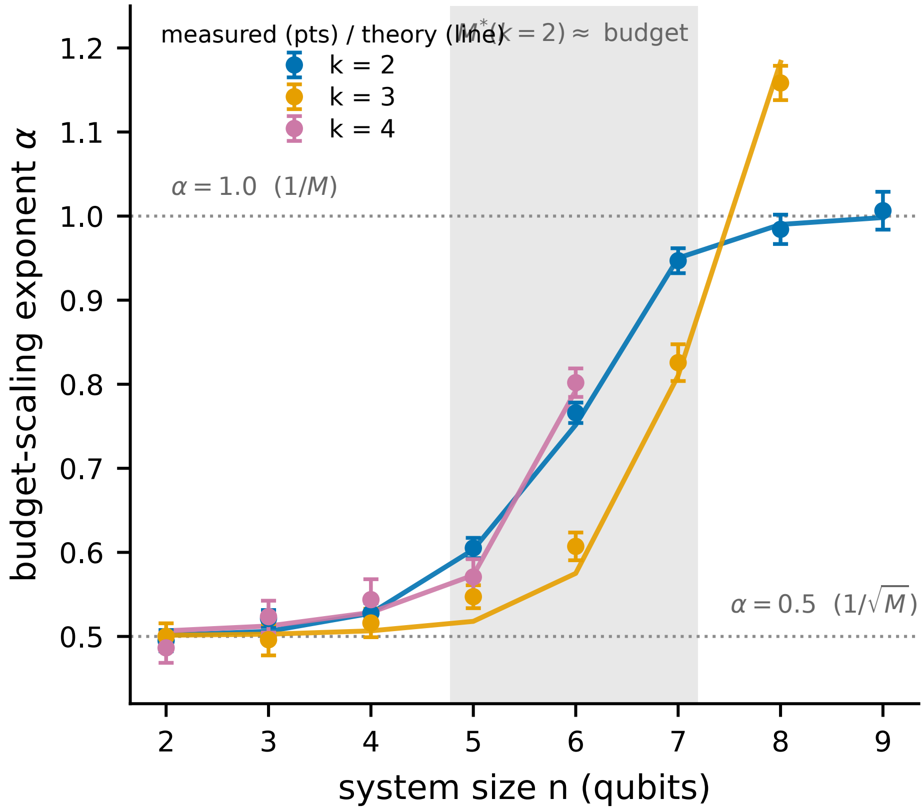

FIG. 1. The effective budget-scaling exponent transition, the paper's central phenomenon. Measured budget-scaling exponent α (RMSE $\propto M^{-\alpha}$, fitted over budgets 2000–128000; markers with bootstrap standard errors) versus system size n for moment orders $k = 2, 3, 4$, with the derived-law prediction (lines). *The prediction curves have no parameters fitted to the plotted points, computed from projection variances fed into the exact variance formula of Sec. 3.1, and are not fits to the plotted points.* As n grows at fixed budget and M falls below the threshold M^*, α migrates continuously from 1/2 toward 1, and past it for the higher moment orders.

and negative through the middle — so the fitted base drifts upward as the window moves out, from 5.14 over $n = 2$ to 7 to 5.60 over $n = 4$ to 9. This is a finite-size fit over the sizes we reach, with the base robust to dropping any single n (it stays within [5.25, 5.49]), not a measurement of an asymptotic rate. Appendix D gives the residual and leave-one-out detail.

3.7. Why a changing exponent matters

A fixed-exponent bound of the form RMSE $\leq C_n/\sqrt{M}$ is not wrong. It is a bound, and it holds. Nor is a fixed exponent what the tighter analyses assume: the two-term treatments keep both orders and exhibit the two regimes directly [18]. What they leave open is where the regimes meet, since bounding ζ_1 and ζ_2 separately does not constrain their ratio, and that is what evaluating the coefficients supplies. The fixed-exponent summary is still worth addressing, because it describes the wrong regime for the systems people are actually running. At nine qubits, with the budgets accessible on current hardware, the estimator sits in the $\alpha = 1$ regime, where additional shots buy error reduction faster than $1/\sqrt{M}$ but from a variance that is already exponentially large. The improved rate does not rescue the protocol, because the exponential prefactor still dominates it. What the transition establishes is that the shape of the sample-complexity curve is not the shape a fixed-exponent analysis assumes, and that any extrapolation of measured scaling to larger systems which presumes $\alpha = 1/2$ will be wrong in a predictable direction. At fixed n the linear term dominates once the budget grows without bound, so the asymptotic scaling remains the familiar square-root law: the migration reported here is a finite-budget effect over the window hardware actually reaches, not a change in the asymptotic sample complexity.

Having characterized the single-copy cost exactly, we turn to the collective route, where the error has a different character entirely.

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4. THE COLLECTIVE ROUTE: TWO EXACT BIAS LAWS

The collective route's error under noise is dominated by a bias rather than a variance. The bias is independent of the shot budget: taking more shots drives the finite-shot term away but leaves the bias untouched, a prediction we test directly in Section 5.3 and which holds. The size of that bias is given exactly by one of two laws, and which law applies is determined by the geometry of the noise channel rather than by its rate.

4.1. Global depolarizing noise: linear in g, with no compounding

Let the kn-qubit collective register be subject to global depolarizing noise at rate g, so that $\rho^{\otimes k} \mapsto (1-g)\rho^{\otimes k} + g\,\mathbb{I}/d^k$ with $d = 2^n$. Then

$$\mathrm{Tr}\big(C_k\,\big[(1-g)\rho^{\otimes k} + g\,\tfrac{\mathbb{I}}{d^k}\big]\big) = (1-g)\,\mathrm{Tr}(\rho^k) + g\,\frac{\mathrm{Tr}(C_k)}{d^k} = (1-g)\,\mathrm{Tr}(\rho^k) + g\,2^{\,n(1-k)},$$

using $\mathrm{Tr}(C_k) = d$ from Section 2.3. Hence

$$\boxed{\text{bias} = g\,\big|\mathrm{Tr}(\rho^k) - 2^{\,n(1-k)}\big|}$$

The bias is linear in g and does not compound across qubits. That result contradicts the natural intuition, which is that noise acting on kn qubits should accumulate as $1-(1-g)^{kn}$. That compounding form overestimates the bias by a factor of five to fourteen. The error is structural rather than numerical: a per-qubit compounding law was applied to a channel that acts globally. The single-cycle property of C_k is what collapses the compounding, and any treatment that ignores the channel's geometry gives the bias a qualitatively different dependence on n: the compounding form saturates toward unity as kn grows, while the true bias stays linear in g and does not compound across qubits, so the two make parametrically different predictions.

4.2. Per-qubit channels: the noise relabels the state

Let $\mathcal{E}$ be any single-qubit channel acting on every qubit of every copy of the register held for the collective test: the noise this route bears and the single-copy route does not, with ρ the reference state and $\mathrm{Tr}(\rho^k)$ the estimand. What makes the quantity below a bias rather than an identity is that $\mathcal{E}$ is specific to this route, and we say so explicitly because it is a modeling choice and not a derivation.

The identity that follows needs one further precondition, and it is narrow: the k copies must be independent and must undergo $\mathcal{E}^{\otimes n}$ *before* an otherwise ideal cyclic-permutation measurement. It does not cover noise inside the cyclic test — imperfect controlled-permutation gates, ancilla decoherence, readout error — nor correlated noise acting across copies, nor any channel acting during or after the circuit. Those perturb the measured observable or the instrument, not only the state being measured, and the relabeling below does not apply to them. Within that precondition, applying the same channel to every qubit of every copy is identical to applying $\mathcal{E}^{\otimes n}$ to each copy independently, so

$$\mathcal{E}^{\otimes nk}\big(\rho^{\otimes k}\big) = \big(\mathcal{E}^{\otimes n}(\rho)\big)^{\otimes k} = \sigma^{\otimes k}, \qquad \sigma \equiv \mathcal{E}^{\otimes n}(\rho),$$

and the cyclic test of a product of identical states returns the k-th moment of that state:

$$\boxed{\text{measured} = \mathrm{Tr}(\sigma^k), \qquad \sigma = \mathcal{E}^{\otimes n}(\rho)}$$

so that the bias is $\big|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)\big|$.

Under a per-qubit channel the noise does not corrupt the measurement in the usual sense; it relabels which state is measured. The cyclic test returns the exact k-th moment of the damaged state $\sigma = \mathcal{E}^{\otimes n}(\rho)$ rather than of the reference ρ, and that substitution is the entire bias.

Two things this does not say. It does not exclude preparation noise, which is already in the model and absorbed into ρ: the noisy-pure ensemble of Section 2.5 is a prepared pure state carrying depolarizing noise at rate q, both routes see it, and it contributes no bias at all. And it is not a claim about when $\mathcal{E}$ acts. A channel common to both routes would bias the single-copy route by the same substitution, since shadows of σ estimate $\mathrm{Tr}(\sigma^k)$ and not $\mathrm{Tr}(\rho^k)$; the substitution would then cancel from the comparison and carry no information about which route is cheaper. What the law requires is that $\mathcal{E}$ be borne by the collective route alone.

This explains a puzzle in the data that a rate-based model cannot. No universal effective attenuation rate fits the measured biases, because the bias is not a rate. It is however much the channel happens to deform that particular state's spectrum, and the two states of identical purity can be deformed by different amounts.

4.3. Verification

Both laws are exact identities rather than approximations, and we verified them accordingly: against explicit construction of C_k and the noise-damaged k-copy state, at $n = 2, 3$ and $k = 2, 3, 4$, for global depolarizing, amplitude damping, and dephasing, agreeing to approximately 10^{-15}. We take the single-qubit Kraus operators at rate g in the standard form: amplitude damping as $K_0 = \mathrm{diag}(1, \sqrt{1-g})$ and $K_1 = \sqrt{g}\,|0\rangle\langle 1|$; dephasing as $K_0 = \sqrt{1-g/2}\,\mathbb{I}$ and $K_1 = \sqrt{g/2}\,Z$; and global depolarizing in the closed form of Section 4.1. Each is applied to every qubit of every copy. They continue to hold at three times the noise rates in which they were derived, and on state ensembles they had never been tested against: Haar-pure ($q = 0$); low-rank, $\rho = GG^\dagger$ with G a $2^n \times 2$ complex Ginibre matrix normalized to unit trace, whose spectrum is not the rank-one-plus-identity form the noisy-pure closed forms assume; and GHZ-noisy, $(1-q)|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}| + q\,\mathbb{I}/2^n$ at $q = 0.15$ with $|\mathrm{GHZ}\rangle = (|0\cdots 0\rangle + |1\cdots 1\rangle)/\sqrt{2}$. An exact identity is not expected to degrade under extrapolation, and it does not.

With the single-copy variance and the collective bias both known exactly, the crossover follows by setting one against the other.

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5. THE CROSSOVER

5.1. The law

The single-copy error grows exponentially in n and falls with the budget. The collective error is a bias floor plus a finite-shot variance: the floor is bounded above by $1 - d^{1-k}$ with $d = 2^n$, and hence by unity, and is independent of the budget, while the shot term vanishes as the budget grows. The crossover size n^* is where the single-copy error rises above the collective error,

$$\underbrace{\sqrt{\mathrm{Var}(U_M)}}_{\text{exponential in } n,\ \text{falls with } M} \quad = \quad \underbrace{\sqrt{\mathrm{bias}(g,k,n)^2 + \frac{1-\mu^2}{N}}}_{\text{bounded floor + shot noise}}$$

Here μ is the noisy collective signal that the cyclic test returns and $N = M/k$ is the number of k-copy measurements a copy budget of M allows. The shot term is the variance of the bounded cyclic-test outcome, which is a two-valued random variable; it falls away as the budget grows, leaving the bias floor as the large-budget limit. This is the quantity our crossover is computed from throughout.

Every quantity on both sides comes from Sections 3 and 4. Nothing is fitted, and the law has no free parameters, in the sense that no quantity is fitted to the data being predicted;

the laws take the state's projection variances and the noise channel as inputs.

The bounded-floor claim is a statement about bias in *estimation* at a fixed additive precision ε: the collective-test outcome is a bounded random variable, so estimating $\mathrm{Tr}(\rho^k)$ to fixed ε costs $O(1/\varepsilon^2)$ shots at every n, and the collective error saturates at the bias floor rather than growing. It is not a claim that collective measurement is cheap for all tasks. For a task whose target precision must itself shrink exponentially in n (purity *testing* of a Haar-random pure state against the maximally mixed state is the canonical case, where the gap to be resolved closes exponentially under noise), [6] proves that exponentially many samples are required even with two copies, and that no adaptive strategy escapes this. Our floor bounds the bias of estimation; it does not make exponentially fine testing cheap. The quantitative reconciliation, including why our operating regime is unaffected, is in Section 7.

5.2. Validation on 83 cells and four ensembles

Across 83 cells spanning moment order $k \in \{2, 3, 4\}$, three noise models (depolarizing, amplitude damping, and dephasing), noise rates from 0 to 0.3, system sizes $n = 2$ to 9, four budget multipliers, and four state ensembles, the criterion predicts 82 of the 83 simulated crossovers within one qubit and 73 of 83 exactly (Figs. 2 and 3). The comparison is asymmetric between the two routes in two respects, and both are deliberate. The first is evidential. The single-copy side of each cell is a forward simulation of the estimator, fresh snapshots reduced by the U-statistic, so the variance law is tested against data it did not generate; the collective side is the exact bias law evaluated analytically plus its shot noise.

The second is the noise model, and it is the stronger assumption. The channel of Section 4 degrades the collective route only; the single-copy route is the ideal, noiseless statistical baseline, on the reading that a joint k-copy measurement is the harder thing to implement cleanly. Real shadow readout is not noiseless, and Section 6.3 finds readout to be the dominant error on the device we tested, so this understates the single-copy cost. It understates it in the direction that costs us: in the cases we checked, adding noise of the kind modelled here to the single-copy route raised its RMSE at these budgets and moved the crossover to smaller n. We report this as an empirical observation for the tested noise models and budgets, not a monotonicity theorem: added noise shifts bias and variance together, and at small budgets it can lower the finite-sample error by reducing variance faster than it

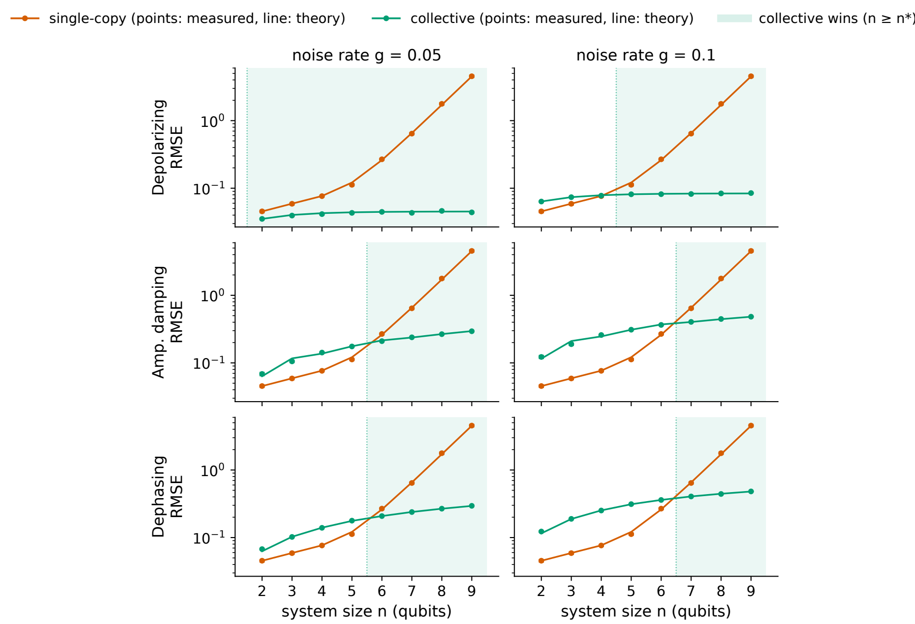

FIG. 2. The crossover for purity ($k = 2$). Single-copy classical-shadow RMSE (vermillion) and collective two-copy RMSE (green) versus system size n at a fixed copy budget, for three noise channels. *The theory curves are predictions with no parameters fitted to these points, from the exact variance and bias laws (Secs. 3–4), not fits to these points.* The crossover n^* is where the exponentially growing single-copy error overtakes the collective error, a bounded bias floor plus a finite-shot variance that vanishes as the budget grows.

adds bias, so the reported sizes are not a guaranteed upper bound. What the asymmetry does shape, and we do not claim otherwise, is the dependence of n^* on the noise rate: that dependence is a property of the collective side by construction. These cells therefore demonstrate consistency with the derived identities, and the out-of-ensemble transfer below is the evidence that the law generalizes. What transfers is the structure of the formula: the projection variances are re-estimated for each ensemble, so this is not zero-shot prediction from universal constants.

Three of the four ensembles, Haar-pure, low-rank, and GHZ-noisy, were never used in deriving the law. GHZ states in particular are maximally non-Haar-typical, and we included them specifically because an ensemble-tuned theory ought to fail on them. The theory's accuracy on GHZ is within a factor of about 1.7 of its accuracy elsewhere, which is the same order of magnitude rather than a breakdown. A law that transferred to structured, low-rank, and pure states without refitting is not a curve fit to the ensemble it was built on.

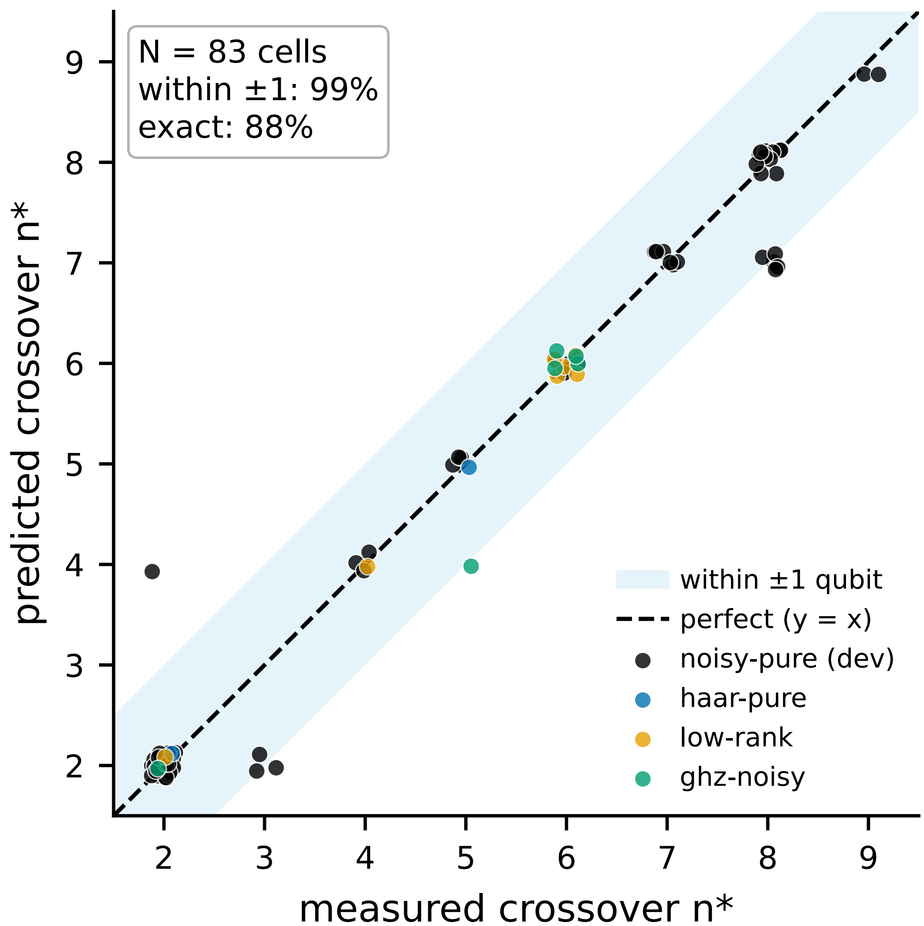

FIG. 3. Crossover validation across 83 cells. Predicted versus Monte-Carlo-observed crossover size n^* over moment orders $k \in \{2, 3, 4\}$, three noise models, and four state ensembles (three unseen by the theory). The criterion places 82 of these 83 simulated crossovers within one qubit of the prediction and 73 of 83 exactly; the cells are simulation, not hardware.

5.3. Three qualitative predictions, and the one we got wrong

Two of the three we predicted correctly in advance; the third we got backwards, and the law corrected us.

Higher noise moves the crossover later on the channels and rates we swept. A larger bias floor therefore takes longer for the exponentially growing single-copy error to climb over.

A larger budget moves the crossover later. The single-copy error falls with budget while the bias floor does not. The collective RMSE plateaus exactly at the predicted floor as the budget grows, while the single-copy RMSE continues falling, a direct test of the bias-versus-variance distinction on which the whole crossover rests, and it holds.

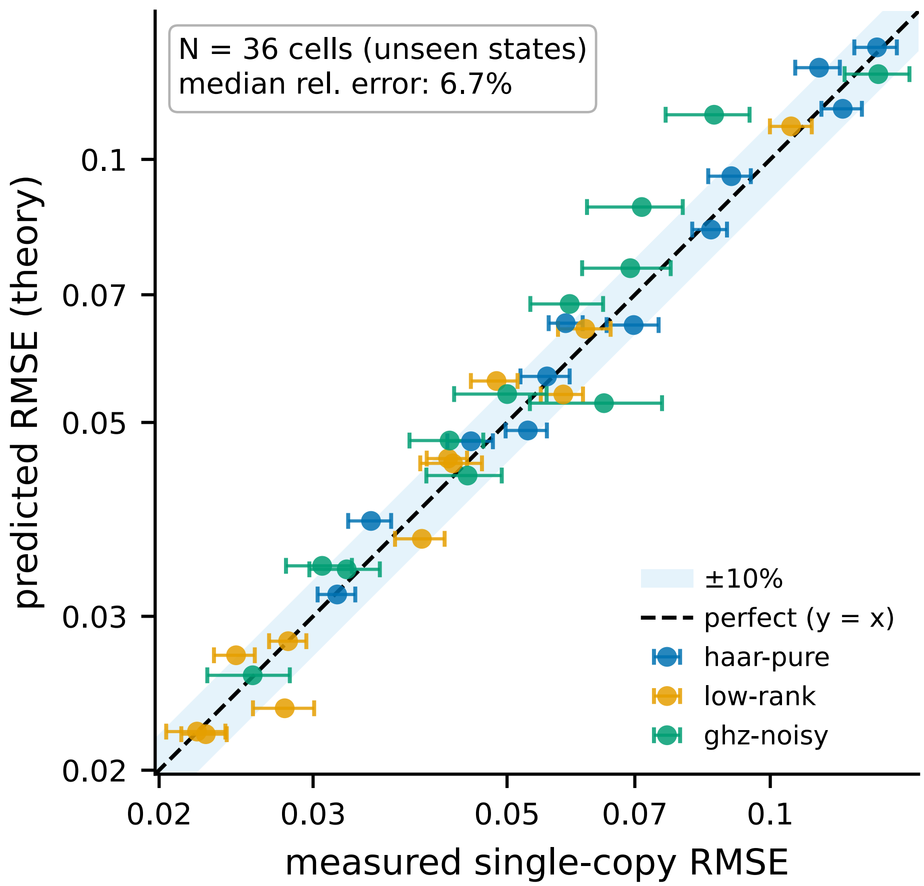

FIG. 4. Out-of-ensemble validation of the variance law. Predicted versus measured single-copy RMSE on three ensembles the theory never saw (Haar-pure, low-rank, GHZ-noisy). The scatter about the diagonal is finite-trial estimation noise (median relative deviation 6.7%; Sec. 8), not a systematic error.

Higher k moves the crossover later on the families we tested. This is counterintuitive: a naive variance argument says that higher moments compound the single-copy variance faster and should therefore cross earlier. The data says the opposite, and the law explains why: $\mathrm{Tr}(\rho^k)$ shrinks with k, so the absolute bias floor that the single-copy error must climb over is lower, and single-copy therefore holds out to larger n values. We report this as an observation about the ensembles and channels swept here rather than as a general rule: $|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)|$ need not fall monotonically with k for an arbitrary spectrum and channel. Purity ($k = 2$) crosses at $n \approx 6$ to 7 under realistic noise, while $k = 3$ and $k = 4$ hold out to $n \approx 8$. The signal-against-a-fixed-floor argument dominates the variance-compounding argument.

5.4. The exponential wall

The practical statement of the result is a single sequence of numbers. Using the exact copy-fair estimator on noisy-pure states at a fixed copy budget, the single-copy purity RMSE runs

$$0.043 \;\to\; 0.072 \;\to\; 0.270 \;\to\; 1.62 \;\to\; 11.98$$

at $n = 2, 4, 6, 8, 10$, growing by a factor of about two per qubit on average and accelerating, to roughly 2.7 per qubit over the last step. The true purity of these states is approximately 0.81. (Fig. 5).

At ten qubits the single-copy estimate carries an error fifteen times larger than the quantity it is estimating. The collective route over the same range stays bounded, because its error is a bias floor and the floor cannot exceed $1 - d^{1-k} < 1$: both $\mathrm{Tr}(\sigma^k)$ and $\mathrm{Tr}(\rho^k)$ lie in $[d^{1-k}, 1]$, so their difference is bounded by the width of that interval. The argument holds for every n, every k, and every channel; the bound itself is not n- and k-free, since d^{1-k} carries both. What is universal is the weaker statement that the floor stays below unity, and that is what the crossover argument uses.

This number speaks directly to the resource-cost concern behind the choice in [5] to keep its measure-first protocol single-copy, though that learning task is not moment estimation, and the open questions it poses are not ours. The entangling overhead that motivates such caution is real, and it is not the binding constraint. The binding constraint is that single-copy estimation of nonlinear functionals by the local-randomized-measurement shadow U-statistic we study does not survive to the system sizes they are entering — a statement about this protocol, not about every single-copy strategy. Whether the collective route survives on a real device is a separate question, and it is the one we take up next.

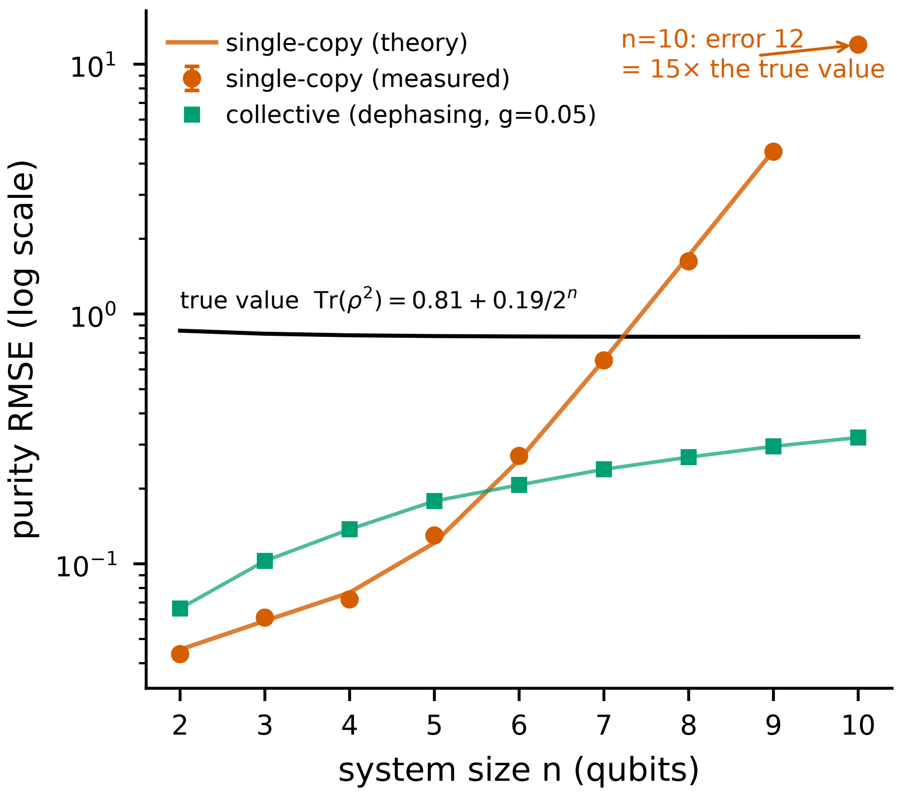

FIG. 5. The exponential wall. Single-copy purity RMSE versus n out to $n = 10$ at a fixed copy budget of $M = 2000$ (log scale; measured points with 68% CIs, theory line with no parameters fitted to the crossover data; noisy-pure ensemble, collective route under dephasing at $g = 0.05$), with the exact per-n true purity $\mathrm{Tr}(\rho^2) = 0.81 + 0.19/2^n$ marked (0.858 at $n = 2$, approaching 0.81). By $n = 10$ the single-copy error reaches ~ 12, about fifteen times the quantity being estimated, while the collective error (green) stays bounded.

6. HARDWARE: A COMMITTED-IN-ADVANCE TEST ON RIGETTI CEPHEUS-1-108Q

6.1. Platform, and a disclosure about the execution path

All measurements were submitted to the Rigetti Cepheus-1-108Q backend as provisioned by the platform, a 108-qubit superconducting processor built from twelve interconnected nine-qubit chiplets. Published median fidelities at the time of the experiments were 99.1% for the two-qubit CZ gate and 99.9% for single-qubit gates. Readout error is not published, a fact that matters (Section 6.3).

Access was obtained through the Open Quantum platform operated by Quantum Rings [9], which provides a unified API to QPUs from multiple vendors. We used the Public Tier throughout. Two properties of that tier are material to the interpretation of our results.

First, the Public Tier does not route jobs directly to the vendor. It is powered by the Quantum Compute subnet (SN48) on the Bittensor network, operated by qBitTensor Labs: circuits are routed to distributed operators who execute them on the target hardware, and validators perform spot checks to confirm that circuits were executed on the appropriate target QPU [9]. We therefore cannot independently verify the execution path of any individual job beyond the platform's own validation. This does not affect Sections 6.2 and 6.3, which are internally consistent characterizations, but it is a candidate contributor to the session-to-session variability reported in Section 6.4 that we are not able to rule out, and we list it as such in Section 8.

Second, use of the Public Tier carries two license conditions: publications resulting from work on the tier are required to cite the platform paper [9], which we do, and circuits, results, and metadata are contributed in anonymized aggregated form to a common repository maintained by Open Quantum [9].

Jobs were submitted as OpenQASM 3 with physical-qubit addressing, which the platform required because it rejects non-contiguous virtual registers.

6.2. The modeled entangling overhead is small for the tested circuits

The destructive SWAP test at $n = 2$ transpiles onto physical qubits $\{0, 1, 9, 10\}$ using four CZ gates and zero routing SWAPs. The GHZ ladders at $n = 3$ and $n = 4$ likewise map with zero routing, at $3n - 2$ CZ gates, and the ladders nest. CZ error on these qubits was not measured directly: the platform compiler resynthesized our characterization echo and optimized its gates away, and the verbatim box that would have prevented this failed at execution. We therefore assume CZ error at the published median of 0.9%. The published figure is an average gate infidelity r; under the depolarizing convention $\Lambda(\rho) = (1-p)\rho + p\,\mathbb{I}/d$ it corresponds to a channel parameter $p = \frac{d}{d-1}r = \frac{4}{3}r = 0.012$ for a two-qubit gate ($d = 4$), the value the deficit decomposition below uses. This is a device-wide median, not an edge-specific measurement, and a depolarizing model does not capture coherent error or leakage, so the gate contribution should be read as a rough scale, not a precise figure. The Bell run supports that assumption indirectly, since reconciling it against the measured readout favors spec-level CZ over zero CZ. On that assumption, gate error accounts for roughly 0.042 of the purity deficit at $n = 2$ against a total deficit of 0.29.

The resource-demand concern cited in [5], that collective measurement demands significant qubit count, two-qubit depth, and routing overhead, does not materialize for the destructive SWAP test on a square-lattice device. Four two-qubit gates and no routing make the entangling overhead a small part of this case under the assumed CZ error: the gates account for about a seventh (roughly 14%) of the observed error under that model. Whatever is degrading the collective measurement on this hardware, it is not the entangling overhead.

By contrast, generic (Haar-random) states at $n = 4$ require 46 CZ gates on this topology — 26 intrinsic to the circuit plus 20 routing-attributable CZ (about seven physical SWAPs' worth) inserted for the coupling map — against the GHZ ladder's 10 CZ with zero routing. The favorable transpilation is a property of structured states on a matched topology, not of the SWAP test in general. We therefore restrict the hardware series to GHZ ladders.

This is a $k = 2$ statement. The ancilla-based Hadamard test required for $k \geq 3$ (Section 2.3) carries a controlled cyclic shift whose cost we have not measured, and the conclusion of this section does not extend to it.

6.3. Consistent with readout and SPAM error dominating, and not in the datasheet

Our locked prediction, computed from the published gate fidelities and an assumed 2% readout error and committed to version control before the job was submitted, was a measured Bell-state purity of 0.8932. The device returned

$$0.7184 \quad [0.6992, 0.7376] \ \ (95\% \text{ bootstrap CI, 5000 shots}),$$

against a true purity of exactly 1. The prediction overshot the measured value by more than seventeen standard errors. The estimand in this section is the target the circuit was written to prepare, so ρ is the ideal Bell state and every deviation from it — gate error, readout error, drift — is charged to the global-depolarizing channel of Section 4.1, which has no single-copy reading to be relabelled into. That is target verification rather than characterization of whatever the device actually produced, and the two coincide only when the preparation is exact.

The observed output structure is consistent with the intended circuit, and the dominant discrepancy is quantitatively accounted for by readout error. The four dominant measured outcomes are exactly the ideal support of the Bell SWAP test, with noise leaking approximately 25% of the weight into the remaining twelve outcomes. This is a correctly executed circuit on a noisier device than we modeled. Inverting the bias law of Section 4.1 gives an effective global depolarizing rate of $g = 0.375$, roughly two and a half times the $g = 0.142$ our locked prediction implied.

A single measurement cannot separate gate error from readout error, because both enter the same number. We therefore measured them separately. Readout characterization, using computational basis states prepared with X gates only and no two-qubit gates, gave the numbers below. What this isolates is the combined state-preparation-and-measurement error — X-gate preparation, relaxation during readout, assignment error, and measurement crosstalk together — not the assignment error alone, so "readout" here should be read as SPAM:

qubit	$P(1 \mid 0)$	$P(0 \mid 1)$
q_0	9.3%	8.9%
q_1	0.7%	7.8%
q_9	6.7%	9.8%
q_{10}	3.2%	6.0%

We write q_N for the physical qubit addressed as `$N` in the platform's OpenQASM 3 syntax. The mean is approximately 6.5% per qubit, roughly three times the value we assumed, and it is asymmetric in the direction expected from T_1 decay during readout. Feeding the measured readout back into the model moves the predicted Bell purity from 0.89 to approximately 0.75, against the measured 0.7184. The device model reproduced the measured purity once the measured readout was supplied, which tests one scalar rather than the whole device model, and the input that was wrong is precisely the one the vendor does not publish.

Readout error on this device is also correlated. Qubit q_0's $P(1|0)$ rises from 1.6% when its neighbors are idle to 16.9% when they are excited. Including this measurement crosstalk closes the residual: the corrected model predicts 0.7163 against a measured 0.7184.

The correlation saturates rather than accumulating. Measured across excitation weights $w = 0, 2, 4, 6$, qubit q_0's $P(1|0)$ runs 0.020, 0.169, 0.242, 0.224: a steep rise from $w = 0$ to $w = 2$, then a plateau. A linear extrapolation from the first two points predicts 0.473 at $w = 6$, overestimating the measured value by 0.25. Because the correlation saturates, the crosstalk must be measured at each excitation weight rather than extrapolated; extrapolating instead would have made the $n = 4$ predictions pessimistic by five points.

6.4. Cross-session variation is the leading explanation for the gap

With readout on the GHZ ladder measured in the same session rather than assumed, and CZ error still taken at the published median, we ran a bracketed same-session experiment (Fig. 6): opening readout calibration, predictions locked and committed to version control from that calibration, SWAP measurements, as well as closing readout calibration. Within-session drift measured approximately 1% per qubit, so the device was stable while the experiment ran.

n	measured	opening band	closing band
2	0.686 [0.672, 0.700]	0.718 – 0.742	0.704 – 0.728
3	0.378 [0.360, 0.397]	0.597 – 0.634	0.593 – 0.631
4	0.420 [0.402, 0.439]	0.506 – 0.553	0.520 – 0.568

All three cells fall below both bands rather than between them, so drift within the session does not account for the gap. The $n = 2$ cell here (0.686, 10,000 shots) is the session-study SWAP test, a different circuit and run from the 0.7184 Bell-purity anchor above, so the two are not repeated readings of one quantity. The degradation is also non-monotonic: $n = 3$ is worse than $n = 4$, despite the $n = 4$ register containing the $n = 3$ register. Neither readout error, which scales with the $2n$ measured qubits, nor CZ count, which grows with n, can produce that ordering.

The cross-session picture explains it. Running the same byte-identical circuits, on the same physical qubits, in different sessions:

Within-session drift is approximately 1% per qubit in readout error, which is not itself a purity. Propagated through the device model it moves the predicted purity by about 0.01 —

register	session 1	session 2	session 3
$n = 3$	0.317	0.378	0.587
$n = 4$	0.431	0.420	0.382

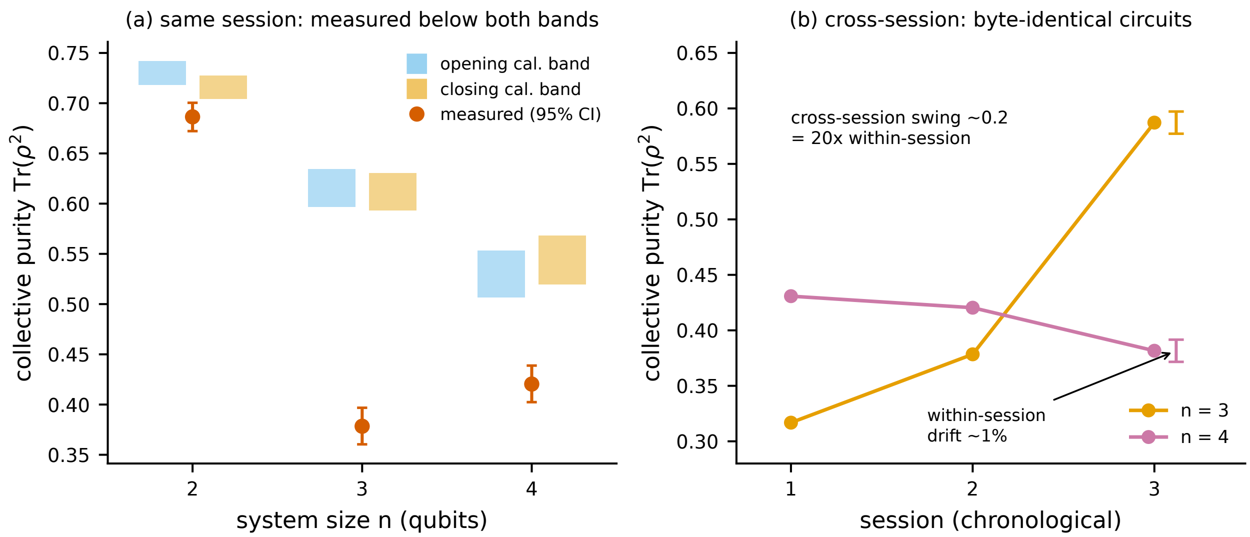

FIG. 6. Hardware: cross-session variation is the leading explanation for the gap. (a) Measured collective purity (vermillion, 95% CI) at $n = 2, 3, 4$ against the same-session prediction bands committed in advance from the opening (blue) and closing (orange) readout calibrations; every measurement falls below both bands. (b) The same byte-identical circuits across three sessions: the $n = 3$ register healed ($0.32 \to 0.59$) while $n = 4$ drifted down ($0.43 \to 0.38$), and the ordering flipped. Cross-session drift ($\sim$0.2) is about 20$\times$ the within-session drift (scale bar).

the gap between the opening and closing bands above. Cross-session drift is approximately 0.2 in purity, roughly twenty times larger on that like-for-like comparison. The $n = 3$ anomaly healed across sessions and the $n = 4$ anomaly persisted, after which the ordering between them flipped. The non-monotonicity that sent us hunting for a physical mechanism was a drift artifact, and identifying it as such required running the same circuit three times over several days, which is not standard practice.

The single-copy route failed the same way. Alongside the collective measurements we ran a fifteen-basis single-copy anchor at $n = 2$, whose U-statistic our shadow pipeline predicted at 1.500 ± 0.028 from the same device parameters. That statistic is not a purity: the fifteen bases collapse to eight distinct unitaries, so about 15% of the snapshot pairs share one, which biases the estimator upward, here past the physical bound $\mathrm{Tr}(\rho^2) \leq 1$. The settings are random Pauli-basis measurements, of which there are nine at $n = 2$, so repeats among fifteen draws are an expected consequence of sampling a finite set rather than a flaw, and the 1.500 prediction is locked for this Pauli-basis pipeline. Because the same estimator is applied to the simulated and the measured data, the anchor tests the pipeline rather than measuring a purity. The anchor samples random Pauli bases while the simulation draws local Haar shadows, but the two ensembles share a second-moment matrix and hence carry the same predicted statistic (Section 3 3.5). The device returned 1.344, with a 95% bootstrap interval of [1.266, 1.433] over 9000 snapshots. The gap is 5.6 times the spread the model predicts for a single run of this anchor, and 3.7 times the wider spread the data themselves show; the intervals are disjoint on either accounting, and no simulated run of the locked model reaches the measured value. We do not adjust the model to fit. Three signals therefore fail together in this session: the collective purities at $n = 3, 4$ fall below their bands, their ordering is non-monotonic, and the single-copy pipeline does not reproduce its own anchor. One common cause accounts for all three, and it is the one this section has been describing: the device parameters were characterized in earlier sessions, and the effective execution channel changed between sessions — byte-identical QASM does not guarantee an identical pulse schedule, and the decentralized route cannot be verified per job. That two protocols with different circuits and different estimators both miss their locked predictions in the same session is what makes drift, rather than any protocol-specific defect, the parsimonious explanation.

6.5. The elimination ledger

Each candidate mechanism was addressed with a quantitative argument rather than by absence of evidence: four are excluded outright, and three are bounded well below the observed deficit.

mechanism	how it was bounded or excluded
Within-session drift	Bracketed calibration measured approximately 1% per qubit, against a 0.2 discrepancy.
Single-copy readout error	Directly characterized. Feeding measured values into the model does not close the gap at $n \geq 3$.

mechanism	how it was bounded or excluded
Gate count	Cannot produce $n = 3$ worse than $n = 4$, since CZ count is monotone in n.
Cross-copy readout crosstalk	Ruled out by a physical bound. Joint readout on the full $2n$ register is factorizable (TVD 0.009 at $n = 3$, 0.011 at $n = 4$), and parity-pair correlations are negligible and no stronger than non-pair correlations. Holding the measured single-qubit flip rates fixed and pushing every copy-pair to its maximum physically realizable correlation, which is bounded because $P(\text{both flip}) \leq$ min of the two marginals and the measured marginals are small, the entire achievable range of predicted purity is $[0.611, 0.669]$ at $n = 3$. The measurement is 0.378. Even maximal correlation leaves 0.23 of the deficit unexplained by this mechanism.
Coherent gate error	Randomized compiling. Pauli-twirled SWAP circuits, with a positive control confirming that the inserted gates physically execute (5.4:1 mass on the disjoint support), moved the purity by -0.013 against the 0.197 same-session $n = 4$ gap, and twirl-to-twirl scatter was shot-noise limited with zero physical scatter. A masking bound places any hidden coherent contribution at no more than 14% of the deficit. The residual is incoherent.

mechanism	how it was bounded or excluded
A specific bad qubit pair ($\{3, 12\}$)	Localization test. An $n = 3$ ladder that includes the suspect pair measured 0.581, matching the standard $n = 3$ ladder's 0.581 to well within the ≈ 0.012 shot-noise standard deviation at 5000 shots, from genuinely distinct raw data. Adding the suspect qubits changes nothing.
A fixed register-geometry fault	The $n = 4$ versus $n = 3$ ordering flips sign across sessions on identical registers. The deficit is session-dependent, not geometric.

6.6. What remains, and what we conclude

What is left is a large, incoherent, session-dependent fault that appears and disappears on the same physical qubits running byte-identical circuits. We do not identify it. Localizing it would require per-edge interleaved randomized benchmarking across many sessions, which is a research program rather than a section, and we do not claim to have done it.

The conclusion the data support is narrower than "we found a new error mechanism" and more useful than "the device is noisy." Static device characterization did not predict collective-measurement performance on this hardware, over the execution path available to us and across the sessions we ran, because the device is not stationary at the relevant scale.

This is not a new observation about NISQ devices in general, and we do not claim it as one. Temporal and spatial instability of superconducting processors is documented: observations collected over 22 months reveal large fluctuations in gate fidelities, duty cycles, and register addressability across temporal and spatial scales, underscoring the limited scales on which NISQ devices may be considered reliable [10, 11], and calibration-drift analyses identify gate error as the dominant cross-device mismatch and readout as secondary [12]. Our contribution is to demonstrate, with committed-in-advance predictions and a seven-mechanism ledger, that this instability is the binding constraint specifically for collective-measurement protocols, at magnitudes that dwarf every error source such protocols are conventionally modeled with. A protocol whose entire advantage rests on a bounded, characterizable bias floor cannot be deployed on a device whose bias floor moves by 0.2 between sessions.

6.7. A practical finding about cloud QPU economics

One further constraint emerged that affects reproducibility rather than physics. The single-copy shadow route requires many independent random measurement bases. On a per-circuit-priced cloud platform without circuit batching, each basis is a separately billable job. An unbiased shadow purity estimate at the precision needed for this comparison requires on the order of a few hundred independent random measurement bases, each a separately billable job – one to two orders of magnitude more than the roughly 3 credits per cell the collective route used at the same shot count.

Single-copy classical shadows are therefore economically infeasible on per-circuit-priced cloud QPUs at the scale required to reproduce their own sample-complexity claims. This is the reason our hardware single-copy baseline at $n \geq 3$ is computed from independently measured device parameters rather than measured directly, with a fifteen-basis $n = 2$ anchor as the one point we could afford to measure. That anchor did not reproduce its prediction (Section 6.4), so the computed points are model output that the hardware did not corroborate, and we present them as such.

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7. RELATED WORK

Classical shadows and variance bounds. The classical-shadow framework [1] gives sample-complexity bounds via the shadow norm. For second-order nonlinear functionals, the variance splits into a kernel term bounded by $4^{|AB|}$ and linear terms bounded by $2^{|AB|}$ [7]. Reference [7] states and applies this bound, citing its own references for it rather than deriving it, so we attribute the statement rather than its origin. These are worst-case, state-independent bounds. For local randomized measurements specifically, Huang, Wang and Xu [21] give worst-case sample-complexity bounds whose exponential base depends on the measurement ensemble — Clifford, local-Haar, and Pauli shadows scale with different constants — for the related task of distributed inner-product estimation; these remain worst-case envelopes rather than evaluated coefficients or a budget threshold. The exact finite-M decomposition of this variance is the standard Hoeffding U-statistic identity [16, 17]; in the classical-shadow setting it is already stated as an equality by [1], and again by Elben et al. [18], who read the two-regime budget scaling off it. Reference [18] bounds ζ_1 and ζ_2 rather than evaluating them. Reference [1] does supply the exact second moment, in its Lemma 4, but applies it to worst-case sample-complexity bounds rather than evaluating the resulting coefficients for a state or an ensemble; neither gives a threshold in M. Separate upper bounds on ζ_1 and ζ_2 constrain their ratio not at all, which is why the threshold does not follow from them. What this work adds is the evaluation of Section 3 3.5, the resulting threshold $M^* = \zeta_2/(2\zeta_1)$, the spectral functionals that fix its growth base, and the comparison against the collective route, which that work does not treat. Alternative shadow ensembles, such as the Hamiltonian-driven shadows of [8], target different observable classes and sample-complexity questions rather than the variance of nonlinear functionals.

U-statistics and the Hoeffding decomposition for quantum moments. Straeter, Tsesmelis and Kwek [13] construct unbiased U-statistic estimators for the partial-transpose moments p_2 and p_3 from randomized homodyne data in continuous-variable systems and derive their variance via a Hoeffding decomposition, obtaining sample-complexity bounds for a p_3-PPT entanglement criterion. This is the closest methodological neighbor to Section 3. The differences are the platform (continuous-variable homodyne rather than qubit local-random-unitary shadows), the target (entanglement detection rather than moment estimation and the collective-measurement trade-off), and the result: their decomposition is exact, as ours is, but the projection variances are bounded rather than evaluated for a state, and no change of scaling regime is considered. We do not claim novelty for the Hoeffding technique itself.

Crossovers in shadow sample complexity. A crossover in shot cost between Pauli and Clifford shadow ensembles for multipartite entanglement witnesses has been reported [14]: ensemble-dependent variance bounds yield qualitatively distinct snapshot-cost scaling, with Pauli-favorable performance for local witnesses crossing over to Clifford-favorable performance as the witness becomes more global. That is a crossover in the choice of measurement ensemble, not between single-copy and collective measurement, and its mechanism is unrelated to ours. The single-copy versus collective comparison has, however, been made on hardware: Peng et al. [19] implement a Fredkin gate and report a reduction in sample complexity for a hybrid two-copy scheme against single-copy shadows. That comparison is empirical and at fixed settings, and is not accompanied by a law locating where the two routes cross.

Statistical-to-bias-floor transitions on hardware. A March 2026 study of shadow tomography on an integrated photonic processor [15] reports that reconstruction error initially follows $O(M^{-1/2})$ in a variance-dominated regime and then saturates at a hardware-determined floor, a transition the authors term a Hardware Horizon. The structure of that argument, statistical error meeting an irreducible hardware bias floor, is adjacent to Section 5, and the two should be distinguished carefully. Their transition is between statistical scaling and a bias floor at fixed system size. Ours is a change in the statistical exponent itself as a function of system size, occurring before any hardware bias enters. Both effects are real and they are not the same effect. The rolling-exponent diagnostic itself is not ours: Aasen and Gärttner [20] fit a power-law exponent to adaptive-tomography infidelity in sliding budget windows and report it drifting from the ideal $1/N$ toward $1/\sqrt{N}$ as the measurement count grows. Their migration is driven by detector noise eroding an adaptive advantage; the one in Section 3.6 is a property of the noiseless estimator's own variance, set by the ratio of the two Hoeffding terms.

Sample-complexity lower bounds for purity. Gong et al. [4] improve the lower bound for purity estimation to $\Omega(\max\{1/\varepsilon^2,\ 2^{n/2}/\varepsilon\})$ for protocols using an identical single-copy projection-valued measurement; their general bound for arbitrary k-qubit-memory protocols is $\Omega(\mathrm{median}\{1/\varepsilon^2,\ 2^{n/2}/\sqrt{\varepsilon},\ 2^{n-k}/\varepsilon^2\})$. Our exact variance is consistent with these bounds and supplies the state-dependent constant they leave open, by evaluating the second moment of [1] over the ensemble.

Collective measurement on noisy hardware. The exponential single-copy versus collective separation is established in [2, 3]. Its fate under noise is the subject of [6] (Cotler, Gong, and Kannan), the closest theoretical work to ours and the one that analyzes the exact primitive we build on. For purity testing, distinguishing an n-qubit Haar-random pure state from the maximally mixed state, given two copies each subject to per-qubit depolarizing noise $D_\lambda(\rho) = (1-\lambda)\rho + \lambda I/2$, their Theorem 2.4 proves a sample-complexity lower bound of

$$\Omega\Big(\min\Big\{\, 2^{n/2},\ \Big(\tfrac{4}{1+3(1-\lambda)^4}\Big)^n \Big\}\Big),$$

and it holds against *any* algorithm, including arbitrary adaptive joint POVMs on the $2n$ qubits with noiseless circuits of any depth after the noise layer. The noiseless task needs $\Omega(2^n)$ samples single-copy and $O(1)$ two-copy; their bound shows the two-copy advantage is degraded by noise by a factor exponential in n, and that no adaptive strategy circumvents it.

The relationship to our result turns on the task. They study purity *testing*: deciding between two hypotheses whose distinguishing gap shrinks exponentially under noise. We study purity *estimation*: additive error ε on $\mathrm{Tr}(\rho^k)$ at a fixed precision. The collective-test outcome is a bounded random variable, so estimation to a fixed ε costs $O(1/\varepsilon^2)$ shots regardless of n. Their exponential cost comes from the testing task requiring an exponentially small ε, the hypothesis gap itself, not from the SWAP test being intrinsically exponential for estimation. We do not evade their lower bound; we solve a different problem.

The two results nonetheless share a mechanism. Our Section 4.2 identity, that under a per-qubit channel the k-copy test returns exactly $\mathrm{Tr}(\sigma^k)$ with $\sigma = \mathcal{E}^{\otimes n}(\rho)$, is the finite-$n$ mechanism underneath their asymptotic theorem for SWAP-based protocols. The collective test faithfully reports the purity of the damaged state σ, and because per-qubit depolarizing drives $\mathrm{Tr}(\sigma^2)$ toward 2^{-n} under *both* the pure and the maximally-mixed hypothesis, the testing gap closes. Feeding our exact law into their setup at $\lambda = 0.1$, the gap $\Delta(n, \lambda) = \mathrm{Tr}(\sigma_{\text{pure}}^2) - 2^{-n}$ falls from 0.54 at $n = 2$ to 0.21 at $n = 10$, and the implied SWAP-test shot count $1/\Delta^2$ rises from 3.5 to 22 over that window. Their bound is asymptotic and leaves its constant open, so the comparison we can make is one of exponential bases rather than of counts at any particular size. The asymptotic base of $1/\Delta^2$, $\big(4/(1 + 3(1-\lambda)^2)\big)^2$, equals their base $4/(1 + 3(1-\lambda)^4)$ at $\lambda = 0$ and exceeds it at finite noise, the difference controlled exactly by $3\big(1 - (1-\lambda)^2\big)^2 \geq 0$; over $n \leq 10$ the large prefactor dominates this small-base exponential, so the growth rate measured on that window sits just below $b(\lambda)$.

Finally, [6] argues explicitly that the asymptotic $n \to \infty$ limit at fixed noise is not the regime relevant to near-term experiments: instance size is effectively fixed by the hardware, the operative question is how performance improves as the noise rate falls, and what is needed is a quantitative condition on λ for a meaningful two-copy advantage at fixed n. Our crossover criterion is exactly that condition. At the rates and sizes of our experiments ($\lambda \approx 0.1$, $n \leq 10$) their lower bound is on the order of tens of samples, from under two at $n = 2$ to about twenty at $n = 10$, so at the scales this paper operates in it imposes no practical obstruction. The bound is a small-base exponential: at $\lambda = 0.1$ its base $b \approx 1.35 < \sqrt{2}$, so the $2^{n/2}$ branch never binds and the cost grows at every n, but it reaches only tens of samples by $n = 10$ and becomes an obstruction only at much larger n. Section 5 quantifies the separation under noise, and Section 6 tests the collective route on hardware.

NISQ device stability. Dasgupta and Humble [10, 11] quantify the temporal and spatial instability of superconducting devices over 22 months, and related work analyzes calibration drift across devices [12]. Section 6.6 confirms their conclusions in the specific setting of collective-measurement protocols and claims no priority over them.

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8. LIMITATIONS

The closed forms are $k = 2$ only, and the prefactor is asymptotic. The second-order projection variances have closed forms for the ensembles considered here, and $M^*(n)$ is analytic for them (Section 3 3.5). Three limits apply. The evaluation uses the Haar moments of a specific ensemble, so for an arbitrary state the identity of [1] must be summed over that state's Pauli spectrum. At $k \geq 3$ we did not obtain closed forms and the coefficients are estimated numerically. And the asymptotic base and prefactor, though exact, are approached slowly: at the sizes reached here $M^*(n)/5.6^n$ is still some fifteen percent below its limit.

The $k = 3$ validation is 6 of 7. The single miss sits on the two-sigma boundary and is sensitive to the random seed. We report it rather than round it to 7 of 7.

The noise model is a channel abstraction. Global depolarizing and independent per-qubit channels are idealizations. The bias laws are exact given the channel, and they say nothing about whether the channel describes any particular device. Section 6 documents exactly where a real device departs from them.

The crossover was not demonstrated on hardware. It sits at $n \approx 5$ to 8 under realistic noise, and device non-stationarity dominates at those sizes on the hardware available to us. Section 6 is a test of the model, not a demonstration of the advantage.

The hardware single-copy baseline at $n \geq 3$ is computed, not measured, for the economic reason given in Section 6.7. The one point we could afford to measure, the $n = 2$ anchor, missed its prediction by 5.6 predicted standard deviations (Section 6.4). The computed points are therefore model output on a device that our own anchor shows the model does not currently track, and nothing in Section 6 should be read as a hardware measurement of the single-copy route.

The execution path on the Public Tier is not independently verifiable by us. Jobs were routed through a decentralized network with validator spot-checking rather than direct vendor access (Section 6.1). We cannot exclude this as a contributor to the session-to-session variability of Section 6.4. It does not affect the readout and gate characterizations, which are internally consistent, but a replication with direct vendor access would be worth running.

Statistical caveats. The out-of-ensemble RMSE validation carries a median relative deviation of 6.7% between predicted and measured values (Fig. 4). We investigated this and found it to be finite-trial noise on both sides rather than a systematic error: the estimator is approximately Gaussian at the budgets used, the projection variances are converged to better than 1%, and the measured RMSE converges to the predicted value to within $\pm 0.3\%$ as the trial count increases. The 6.7% figure matches the intrinsic RMSE noise at the trial counts used, and the signed deviation is $+3.4\%$, which is within noise once cell correlations are accounted for. The predicted crossover n^*_{pred} is reported as an integer throughout, and we do not propagate the projection-variance estimation uncertainty onto it. Resampling the projection variances by their committed relative spread shifts the predicted crossover for 5 of the 60 noisy-pure cells (mean per-cell probability 1.7%), always by exactly one qubit and never by more.

9. CONCLUSION

We began with a number: at ten qubits, on noisy-pure states at a fixed budget, single-copy estimation of purity returns an error fifteen times the quantity being estimated. That number is not an accident of implementation, and this paper explains where it comes from.

That the effective budget-scaling exponent for shadow-based moment estimation is not a constant, and changes character with the measurement budget, is established [18, 20]. What does not follow from the bounds is where it changes: separate upper bounds on ζ_1 and ζ_2 constrain their ratio not at all, so the threshold is not located by them. We evaluate the projection variances exactly for depolarized Haar-random pure states at a single noise rate, verify the resulting variance against brute-force Monte Carlo of the exact estimator at three moment orders and three state ensembles, and confirm its prediction of the exponent out of sample with no fitted parameters. That evaluation fixes the threshold, and with it the exponent over the budget window an experiment can actually reach. The asymptotic scaling at fixed system size is not what moves: that remains the square-root law. Because the fitted threshold grows steeply with n over this range, that window is where the experiments at the sizes we reach sit. The mechanism is a competition between two terms in the estimator's variance whose ratio grows approximately exponentially over the tested range, and locating it is what the numerically evaluated coefficients buy.

Paired with two exact bias laws for the collective route, distinguished by the geometry of the noise channel rather than by its rate, the variance law predicts without free parameters the system size at which collective measurement attains the lower RMSE at equal copy budget. It places 82 of the 83 simulated crossovers within one qubit.

The practical guidance follows. For estimating purity of noisy-pure states under per-qubit amplitude-damping or dephasing noise at the rates and copy budgets tested here, single-copy classical shadows — the exact copy-fair U-statistic on local Haar-random single-qubit shadows, judged by absolute RMSE against a target of order one — stop being the lower-RMSE route at equal copy budget beyond roughly six to seven qubits; the higher moments hold out to about eight, and under global depolarizing noise the crossover falls earlier still. In every case the law says which side of the boundary a given experiment is on. Collective measurement does work in principle, and the entangling overhead that has made the field cautious about it is, for these structured $n = 2$ circuits on matched topology and under the assumed median CZ error, a minor contribution: the two-copy destructive SWAP test adds two two-qubit gates of measurement overhead on top of the two that prepare the copies, with no routing, and under that error model the four together account for about a seventh of the observed error.

What blocks the demonstration is neither variance nor gate error. It is readout fidelity, which on the device we used is three times the value we assumed and strongly correlated with neighboring excitations, and above all it is device stability. Cross-session drift on byte-identical circuits exceeded the within-session drift by a factor of twenty, exceeded every modeled error source, and could not be attributed to any of seven candidate mechanisms, four excluded outright and three bounded well below the deficit. A protocol whose advantage rests on a bounded, characterizable bias floor cannot be fielded on a device whose floor moves by 0.2 between sessions.

That is the actionable conclusion for hardware developers, and it is more specific than a call for better gates. Under the assumed median-CZ model, two-qubit gate error came out smaller than the observed deficit; the figure of merit the field advertises was not what limited this test. Readout fidelity, which vendors do not publish, and run-to-run stability, which vendors do not characterize, are what stand between current devices and a collective-measurement advantage that the criterion places near six qubits under the noise we model.

Two questions follow from this work. The first of the three we opened, whether ζ_1 admits a closed form richer than Pauli weight, is settled at second order in Section 3 3.5: the cubic identity of [1], evaluated over the ensemble, gives ζ_1 and ζ_2 in closed form and $M^*(n)$ analytically, with growth base 28/5 for spread Pauli spectra and 5 for stabilizer-structured pure states. It remains open at $k \geq 3$. Whether the same variance law, applied to the partial-transpose moments, gives an exact sample-complexity account of entanglement negativity estimation, which is the application that motivated us. And whether the drift we measured is a property of this device, this platform, or this class of hardware, which requires a multi-session, multi-device study that we did not have the access to run.

ACKNOWLEDGEMENTS

Quantum hardware access was provided by the Open Quantum platform operated by Quantum Rings Inc. [9], on the Public Tier, whose license conditions require citation of the platform paper and contribution of anonymized aggregated circuit data to a common repository. All hardware experiments ran on the backend reported by the platform as Rigetti Computing's Cepheus-1-108Q processor. [Mentor acknowledgement.]

DATA AND CODE AVAILABILITY

All code, raw measurement counts, and analysis are available at `https://github.com/alex-messerlian/shadow-moment-crossover`. Raw hardware counts were committed to version control before any analysis was performed, and every locked prediction was committed before the corresponding measurement was submitted.

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Appendix A: The exact fourth-moment U-statistic

The full U-statistic for $\mathrm{Tr}(\rho^4)$ requires the sum over all distinct ordered 4-tuples of snapshots,

$$S = \sum_{(i,j,k,l)\ \text{pairwise distinct}} \mathrm{Tr}(G_i G_j G_k G_l).$$

Because the snapshots do not commute, this does not reduce to matrix power sums. It is obtained by Möbius inversion over the partition lattice of the four cyclic slots.

For each set partition P of $\{0, 1, 2, 3\}$, let $T(P)$ be the sum of $\mathrm{Tr}(G_{a_0} G_{a_1} G_{a_2} G_{a_3})$ over all index assignments in which slots in the same block share an index and different blocks are unconstrained. The Möbius function of the partition lattice is

$$\mu(P) = \prod_{b \in P} (-1)^{|b|-1} (|b| - 1)!,$$

and the all-distinct sum is

$$S = \sum_P \mu(P)\, T(P),$$

summed over the 15 set partitions of four elements.

Most $T(P)$ reduce to matrix power sums built from $S_1 = \sum_i G_i$, $P_2 = \sum_i G_i^2$, $P_3 = \sum_i G_i^3$, and $P_4 = \sum_i G_i^4$. The exception is the alternating partition $\{0, 2\}\{1, 3\}$, which requires

$$A = \sum_{i,j} \mathrm{Tr}(G_i G_j G_i G_j),$$

and this has no power-sum representation. It is computed exactly by a tensor contraction. Define the $d^2 \times d^2$ matrix

$$X_{(a,b),(c,d)} = \sum_i (G_i)_{ab}(G_i)_{cd} = \sum_i \mathrm{vec}(G_i)\,\mathrm{vec}(G_i)^\top,$$

reshape it to X_{abcd}, and contract

$$A = \sum_{a,b,c,d} X_{abcd}\, X_{bcda}.$$

Forming X costs $\Theta(Md^4)$ in time, since each of the M rank-one updates touches all d^4 entries, and $\Theta(d^4)$ in memory; the two contractions cost $\Theta(d^4)$. The construction as written is therefore $\Theta(Md^4)$. The estimator used for the results reported here never forms X: it evaluates the alternating term through the per-qubit factorization $\mathrm{Tr}(G_iG_jG_iG_j) = \prod_q \mathrm{Tr}(G_i^{(q)}G_j^{(q)}G_i^{(q)}G_j^{(q)})$ as an $M \times M$ contraction, which costs $O(M^2n + Mnd^2)$ and forms no d^4 tensor. Both routes agree, and the complete $k = 4$ estimator matches brute-force enumeration over all distinct 4-tuples to $\lesssim 10^{-13}$.

Appendix B: The destructive SWAP test

Two copies of an n-qubit state occupy qubits $0 \ldots n-1$ (copy A) and $n \ldots 2n-1$ (copy B). For each $i \in \{0, \ldots, n-1\}$, apply a CNOT with control i and target $i+n$, then a Hadamard on qubit i. Measure all $2n$ qubits. For each measured bitstring, let a_i and b_i be the outcomes on qubits i and $i+n$. The purity is

$$\mathrm{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{i\,:\,a_i=b_i=1\}}\, P(\text{bitstring}).$$

The circuit uses exactly n two-qubit gates, no ancilla, and has depth two after state preparation. The sign rule is invariant under bitstring reversal, so the endianness convention of the backend does not affect the result. We verified the rule against exact simulation for pure and mixed states to 10^{-16}.

Appendix C: Hardware protocol

Platform. Open Quantum (Quantum Rings Inc.) [9], Public Tier, Standard queue, the backend reported by the platform as Rigetti Cepheus-1-108Q. Jobs were submitted as OpenQASM 3 with physical-qubit addressing (`$0`, `$1`, and so on), which the platform required because it rejects non-contiguous virtual registers. The Public Tier routes jobs through the Quantum Compute subnet (SN48) on the Bittensor network, operated by qBitTensor Labs, with validator spot-checking of execution on the target QPU [9]; see Section 6.1 for the interpretive consequences.

Registers. GHZ ladders, verified by re-transpilation to require zero routing SWAPs:

n	physical qubits	CZ gates
2	$\{0, 1, 9, 10\}$	4
3	$\{0, 1, 2, 9, 10, 11\}$	7
4	$\{0, 1, 2, 3, 9, 10, 11, 12\}$	10

The ladders nest, so a single readout characterization on the $n = 4$ register covers all three.

Discipline. Every prediction was computed, printed, and committed to version control before the corresponding measurement job was submitted. Raw counts were committed verbatim before any analysis. Where a prediction failed, the model was not adjusted to fit. Every hardware job was quote-gated against a hard credit ceiling before charging.

Total hardware expenditure. Approximately 115 platform credits across nine experimental campaigns.

Appendix D: Reproducibility and statistics

This appendix records the sampling design behind every headline number, so that the agreement between prediction and measurement can be judged against how the two were generated. All counts are read from the committed generating code; the sizes and seeds below are those actually used.

State and snapshot counts. The projection variances ζ_c that feed the theory curves are estimated from 4 independently drawn states per (n, k), each from 120,000 shadow snapshots (`results/theory_zetas.json`: `n_states` $= 4$, `n_samples` $= 120{,}000$). At $k = 2$ these coefficients are now computed in closed form (Section 3 3.5) and carry no sampling error; the sampled values are retained in the repository, and substituting them changes no crossover prediction and no α verdict. The measured α points and the noisy-pure crossover cells are averaged over 48 independently drawn states per size, 8 Monte Carlo trials each (`run_budget_sweep.py`: `N_STATES` $= 48$, `N_TRIALS` $= 8$). The held-out stress ensembles use 4 states for the variance estimate and 24 for the measurement on the random families (Haar-pure, low-rank), and a single fixed state with 36 trials for the deterministic GHZ-noisy family (`run_stress_test.py`); their variance estimates use 60,000 snapshots. All states are depolarized Haar-random pure states at $q = 0.1$, except the low-rank and GHZ-noisy families defined in Section 4.3.

Train/test separation. The prediction and the measurement it is compared against do not share snapshots. The ζ_c are estimated from one pseudorandom substream (seeded $[\,0, n, s, k, 3\,]$) and the measured RMSE from a disjoint one (seeded $[\,0, n, s, 1, k, t\,]$), so the shot noise in the theory curve and the shot noise in the measured points are independent by construction. The theory curve is never fit to the measured α: it is the exact-variance RMSE evaluated at the estimated ζ_c and fit over the budget grid on its own (`figures/data.py`: the components are read from `theory_zetas.json` and "not re-estimated"). The state draws do overlap: the four states used for ζ_c are, at this seed, the first four of the forty-eight used for the measured points, since both draw ρ from `noisy_pure`$(n, 0.1, \mathrm{rng}[0, n, s, 0])$. The agreement is therefore not built in through shared shots, though it is not a fully held-out state set either.

Bootstrap resampling units. The α standard errors resample the state as the independent unit (2000 resamples, each carrying its whole correlated budget vector; `fit_budget_exponent_bootstrap`), which is the honest unit because the nested budgets share a draw. The hardware confidence intervals resample differently and less cleanly: the single-copy anchor resamples the 9000 snapshots (300 resamples, "within the fixed 15 bases"), and the collective and Bell-SWAP intervals resample per-shot ± 1 values (4000 and 5000 resamples). Shots within one circuit and one calibration are not independent draws of the estimator, so these hardware intervals capture shot noise conditional on the realized circuit and calibration, not the full run-to-run variability; they should be read as lower bounds on the true uncertainty.

The α regression. Both the measured and the theory α are the negative ordinary-least-squares slope of $\log \mathrm{RMSE}$ against $\log M$ over the same per-cell budget grid ($k = 2$: 2000–128,000, capped to 2000–32,000 for $n \geq 7$; $k = 3$: 2000–32,000; $k = 4$: 500–8000), with $\mathrm{RMSE}(M) = \sqrt{\mathrm{MSE}}$ averaged over states. The two differ only in their input — measured MSE versus the exact-variance RMSE — and are otherwise the identical procedure.

Crossover definition. The crossover n^* is the smallest size from which the single-copy RMSE exceeds the collective RMSE and stays above it for the remainder of the swept range (the sustained crossover of `predict_crossover`). It is an integer from the swept sizes; no interpolation is used, and a cell whose curves do not cross inside the tested range is censored (returns none) and excluded from the aggregate rather than counted as a hit or a miss.

Fit uncertainty. The exponential-fit residuals in Section 3.6 are positive at both ends of the range ($+0.20$ at $n = 2$, $+0.13$ at $n = 9$ in $\log M^*$) and negative through the middle, the signature of mild upward curvature; the fitted base accordingly rises with the window, 5.14 over $n = 2$–7, 5.34 over 2–9, 5.60 over 4–9. Refitting with each n dropped in turn moves the base within $[5.25, 5.49]$, with the low-n point exerting the most leverage (dropping $n = 2$ raises it to 5.49). The base is a finite-size fit over the sizes reached, not an estimate of an asymptotic rate.

The full crossover table. The 83 resolved cells break down as: 68 noisy-pure cells drawn from two independent sweeps (`budget_scaling.json` and `moment_sweep_corrected.json`), which comprise 54 distinct $(k, \mathrm{channel}, \mathrm{rate}, \mathrm{budget})$ configurations with 14 evaluated in both sweeps — and all 14 replications agree on both the measured and the predicted n^* — plus 15 cells across the three held-out ensembles. Table II lists all 83. One cell misses by more than one qubit (noisy-pure, $k = 4$, predicted 4 against measured 2), giving $82/83 = 98.8\%$ within one qubit and $73/83 = 88.0\%$ exact. Stratified by the measured crossover, all 52 cells with $n^*_{\mathrm{meas}} \geq 3$ fall within one qubit and 43 are exact; the one miss beyond a qubit sits in the 31-cell $n^*_{\mathrm{meas}} = 2$ group. Counted as configurations rather than evaluations, the 83 cells are 69 independent settings — 54 distinct noisy-pure plus 15 held-out — with 14 noisy-pure replications.

TABLE II: All 83 resolved crossover cells. “mult” is the copy budget as a multiple of the 2000 baseline. A cell counts as within one qubit (✓) when $|\, n^*_{\text{pred}} - n^*_{\text{meas}} \,| \leq 1$.

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	2	amp. damp.	0.00	1	2	2	✓
noisy-pure	2	amp. damp.	0.02	1	2	2	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	4	8	8	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	4	9	9	✓
noisy-pure	2	dephas.	0.00	1	2	2	✓
noisy-pure	2	dephas.	0.02	1	2	2	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	4	8	8	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓
noisy-pure	2	dephas.	0.10	4	9	9	✓
noisy-pure	2	depol.	0.00	1	2	2	✓
noisy-pure	2	depol.	0.02	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	4	5	5	✓
noisy-pure	2	depol.	0.05	16	7	7	✓
noisy-pure	2	depol.	0.10	1	5	5	✓
noisy-pure	2	depol.	0.10	1	5	5	✓
noisy-pure	2	depol.	0.10	4	7	7	✓
noisy-pure	2	depol.	0.10	16	8	8	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	3	amp. damp.	0.00	1	2	2	✓
noisy-pure	3	amp. damp.	0.02	1	2	2	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.00	1	2	2	✓
noisy-pure	3	dephas.	0.02	1	2	2	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	depol.	0.00	1	2	2	✓
noisy-pure	3	depol.	0.02	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	4	4	4	✓
noisy-pure	3	depol.	0.05	16	8	8	✓
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	4	8	7	✓
noisy-pure	4	amp. damp.	0.00	1	2	2	✓
noisy-pure	4	amp. damp.	0.02	1	2	2	✓
noisy-pure	4	amp. damp.	0.05	0.25	2	4	×
noisy-pure	4	amp. damp.	0.05	1	8	8	✓
noisy-pure	4	amp. damp.	0.10	0.25	6	6	✓
noisy-pure	4	amp. damp.	0.10	1	8	8	✓
noisy-pure	4	dephas.	0.00	1	2	2	✓
noisy-pure	4	dephas.	0.02	1	2	2	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	4	dephas.	0.05	0.25	2	2	✓
noisy-pure	4	dephas.	0.05	1	8	8	✓
noisy-pure	4	dephas.	0.10	0.25	6	6	✓
noisy-pure	4	dephas.	0.10	1	8	8	✓
noisy-pure	4	depol.	0.00	1	2	2	✓
noisy-pure	4	depol.	0.02	1	2	2	✓
noisy-pure	4	depol.	0.05	0.25	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	4	3	2	✓
noisy-pure	4	depol.	0.10	0.25	2	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
haar-pure	2	depol.	0.05	1	2	2	✓
haar-pure	2	depol.	0.10	1	5	5	✓
low-rank	2	amp. damp.	0.05	1	6	6	✓
low-rank	2	amp. damp.	0.10	1	6	6	✓
low-rank	2	dephas.	0.05	1	6	6	✓
low-rank	2	dephas.	0.10	1	6	6	✓
low-rank	2	depol.	0.05	1	2	2	✓
low-rank	2	depol.	0.10	1	4	4	✓
low-rank	2	depol.	0.30	1	6	6	✓
ghz-noisy	2	amp. damp.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.10	1	6	6	✓
ghz-noisy	2	depol.	0.05	1	2	2	✓
ghz-noisy	2	depol.	0.10	1	5	4	✓
ghz-noisy	2	depol.	0.30	1	6	6	✓